

BFS Traversal

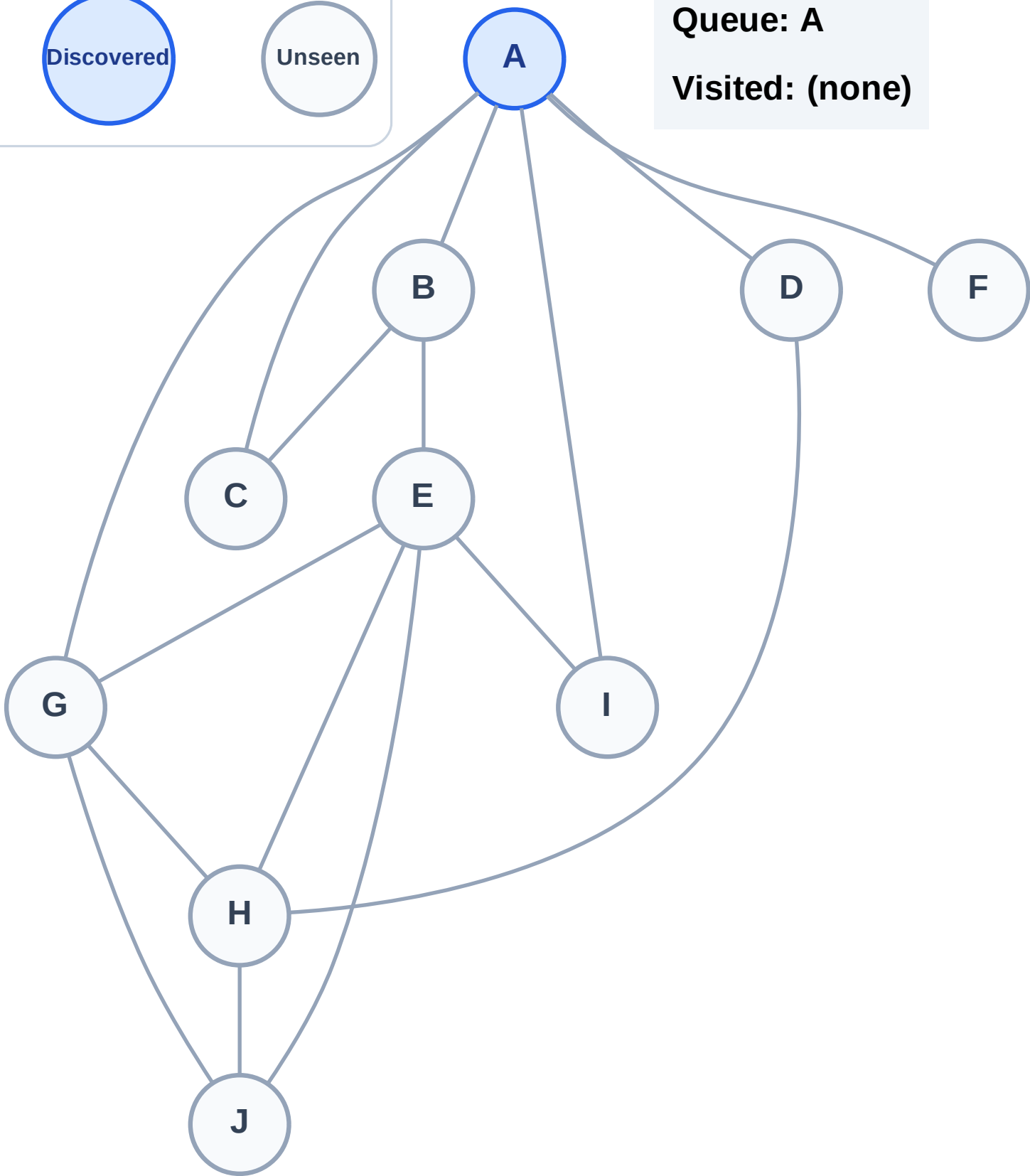
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

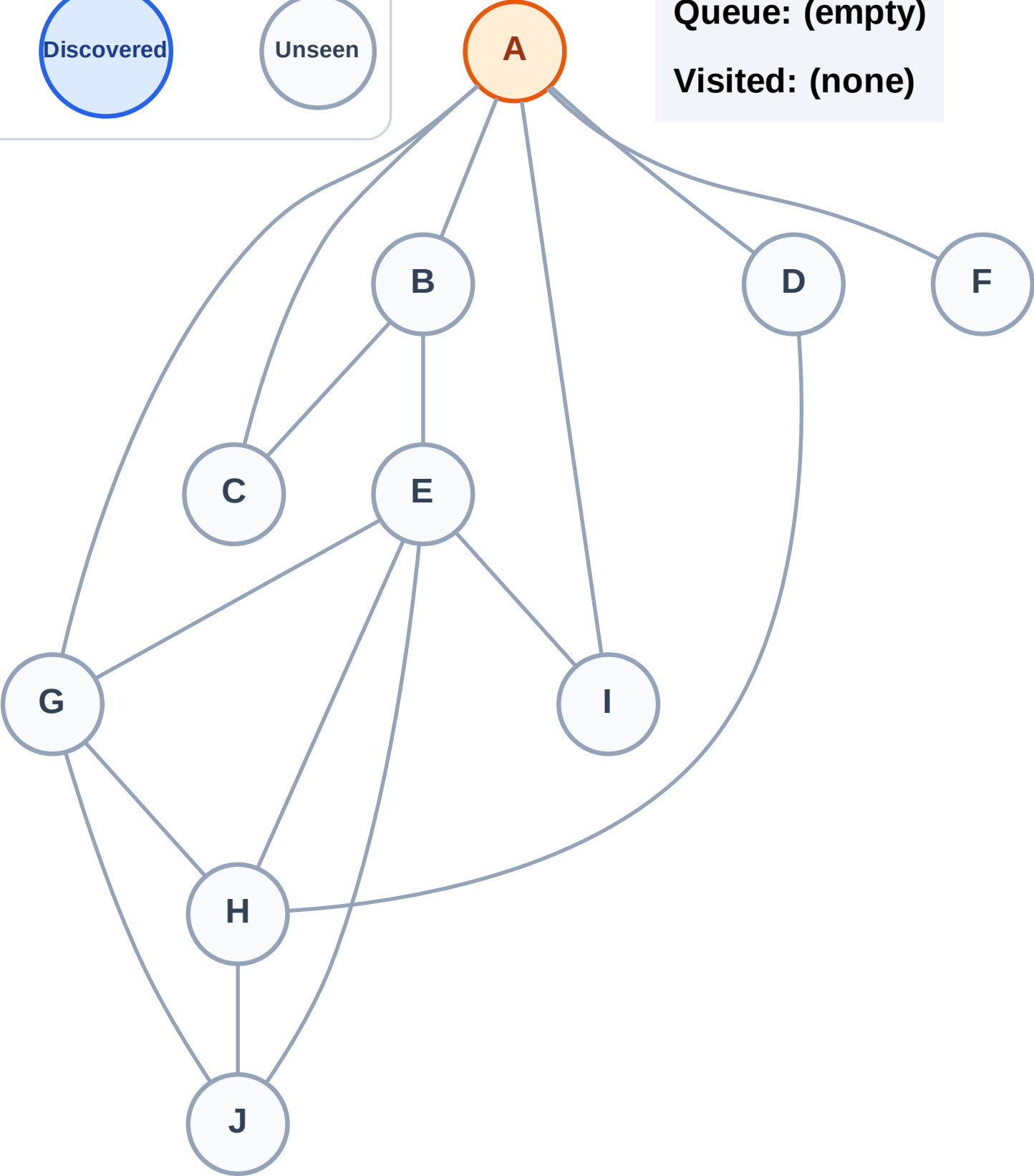
Complete

Processing

Discovered

Unseen

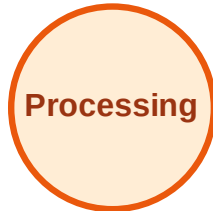
Queue: (empty)
Visited: (none)



BFS Traversal

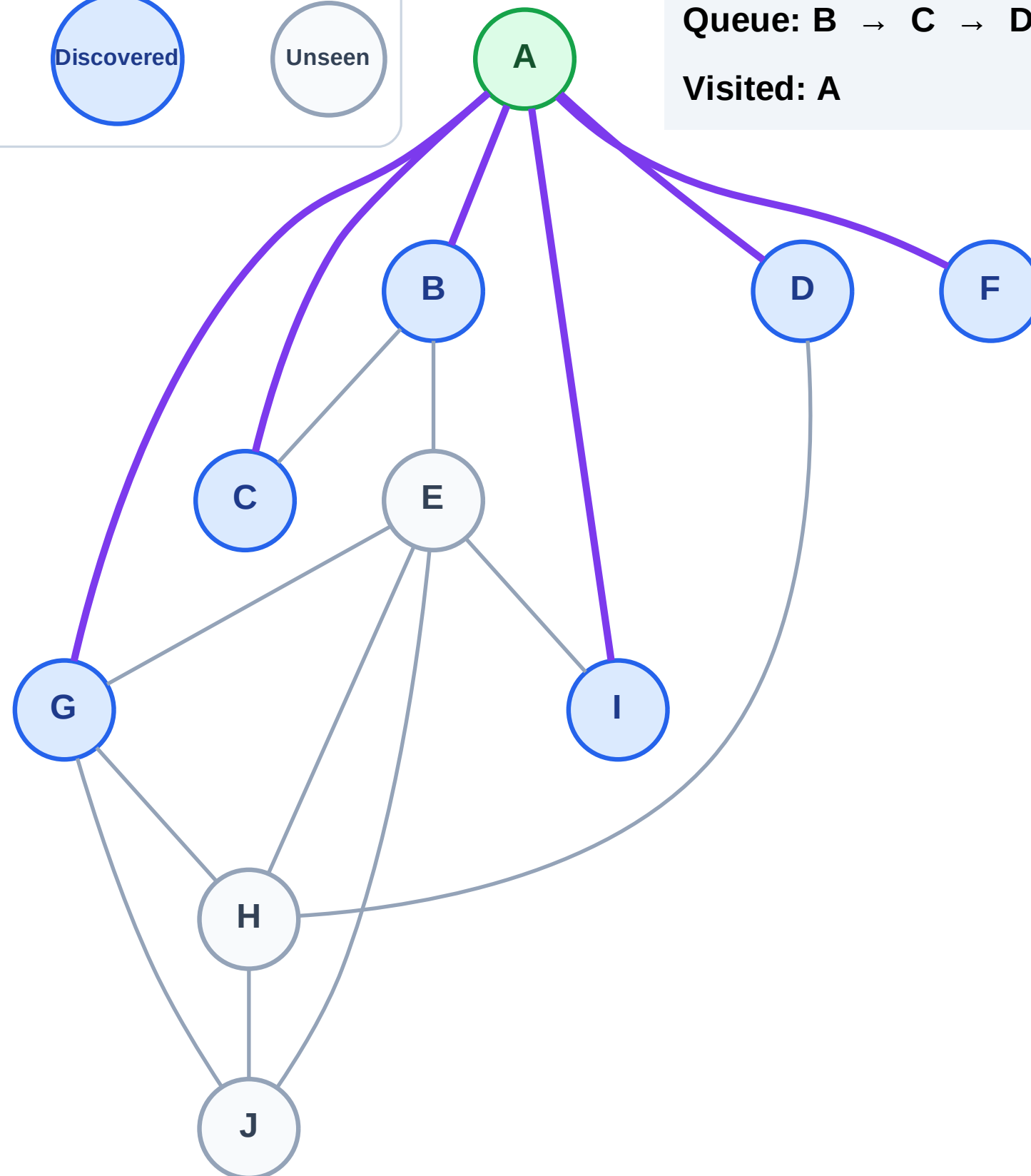
Step 3: Discover B, C, D, F, G, I from A

Legend



Queue: B → C → D → F → G → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

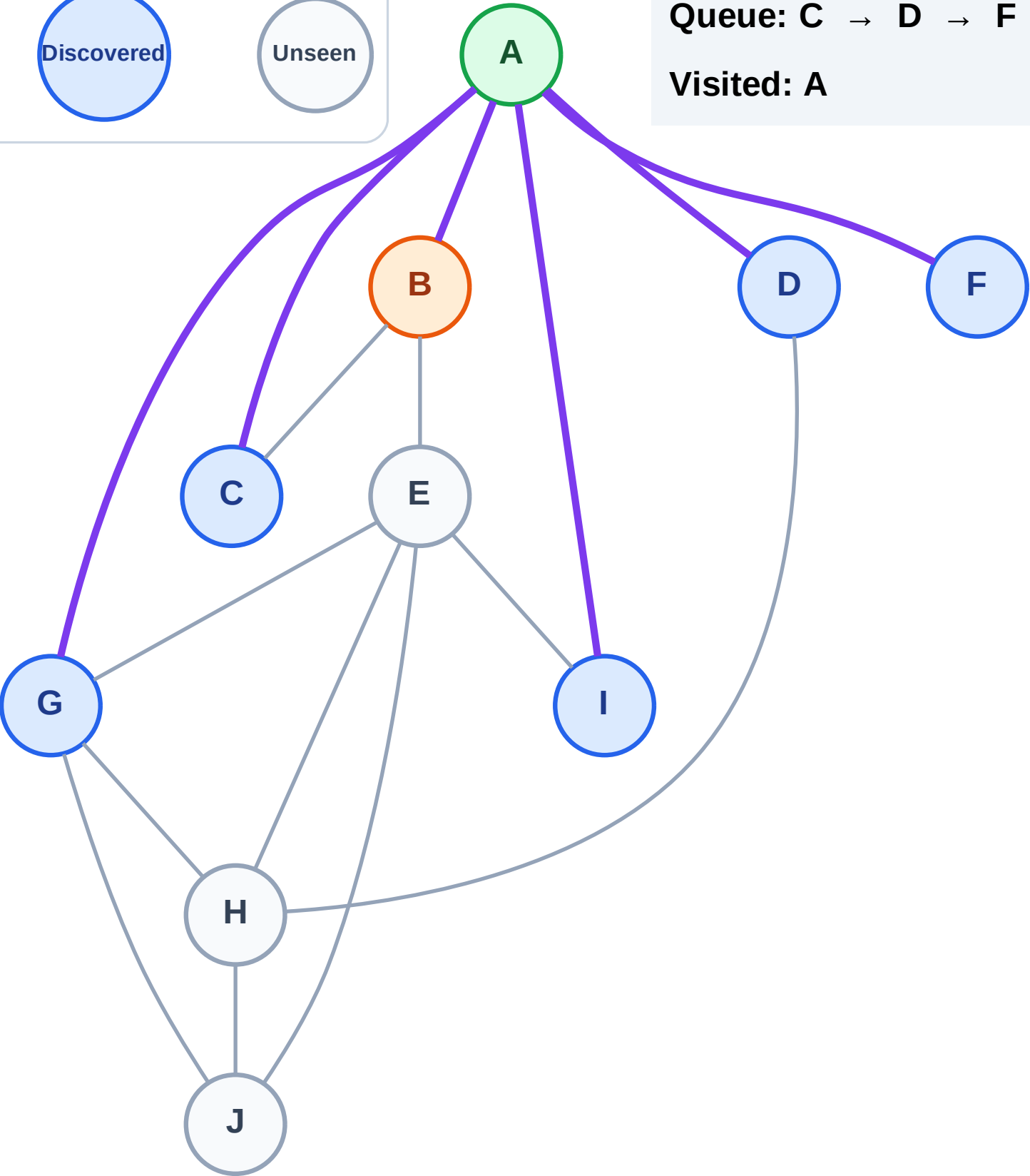
Processing

Discovered

Unseen

Queue: C → D → F → G → I

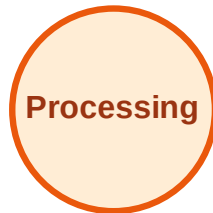
Visited: A



BFS Traversal

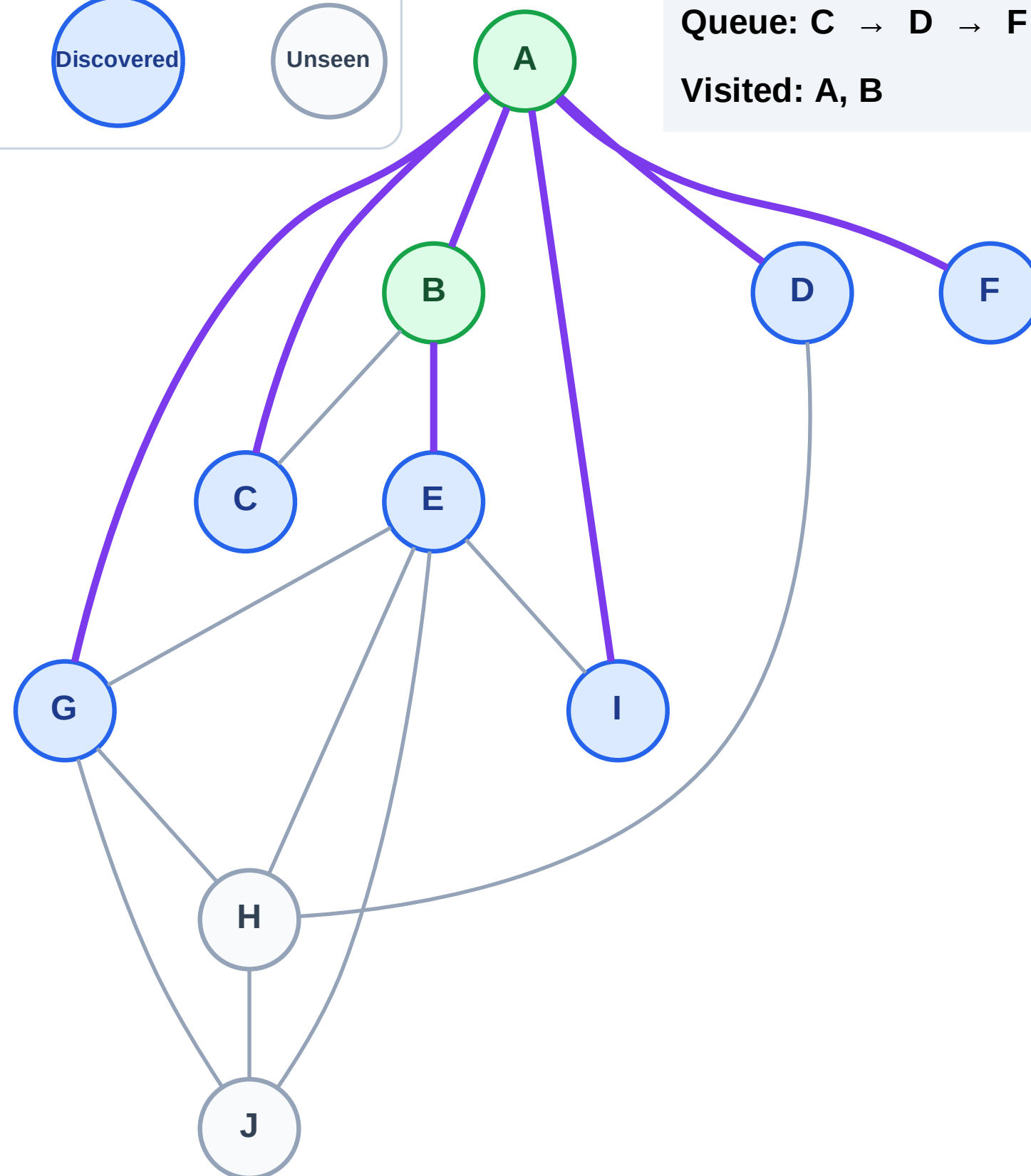
Step 5: Discover E from B

Legend



Queue: C → D → F → G → I → E

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

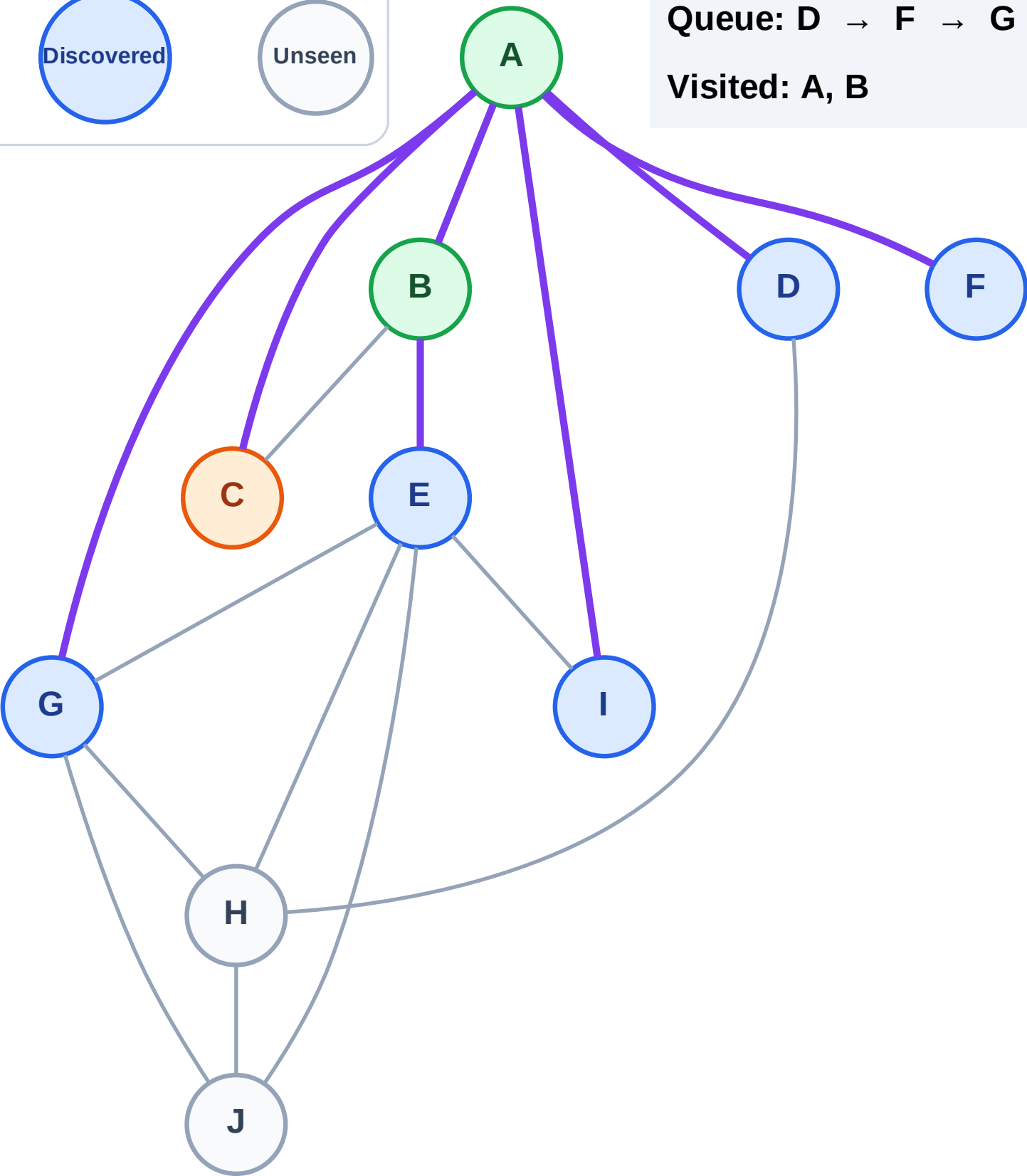
Processing

Discovered

Unseen

Queue: D → F → G → I → E

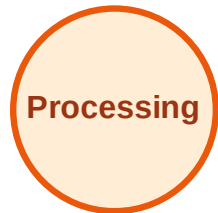
Visited: A, B



BFS Traversal

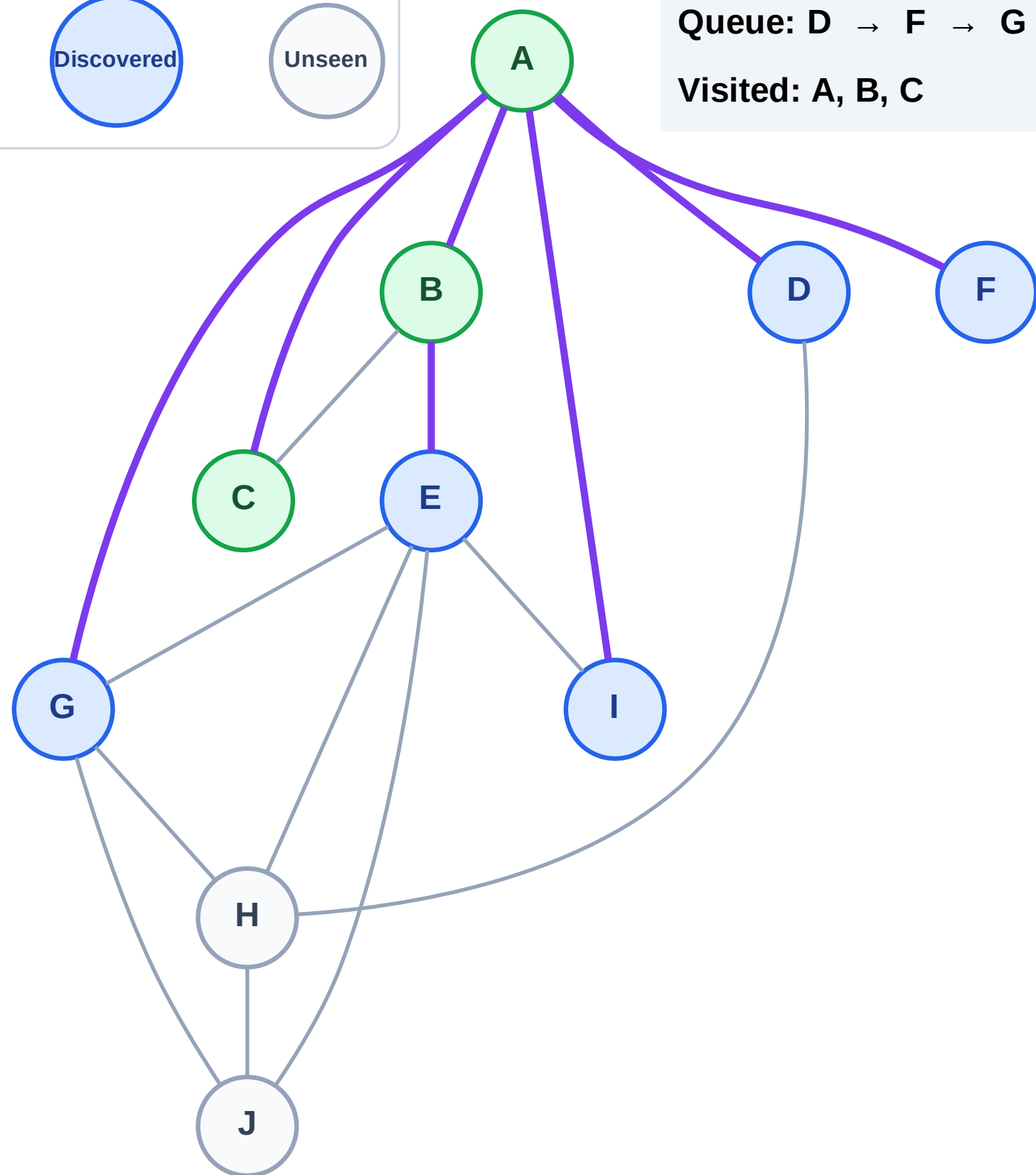
Step 7: No new neighbors from C

Legend



Queue: D → F → G → I → E

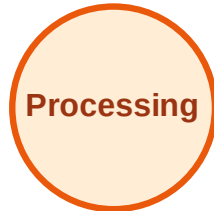
Visited: A, B, C



BFS Traversal

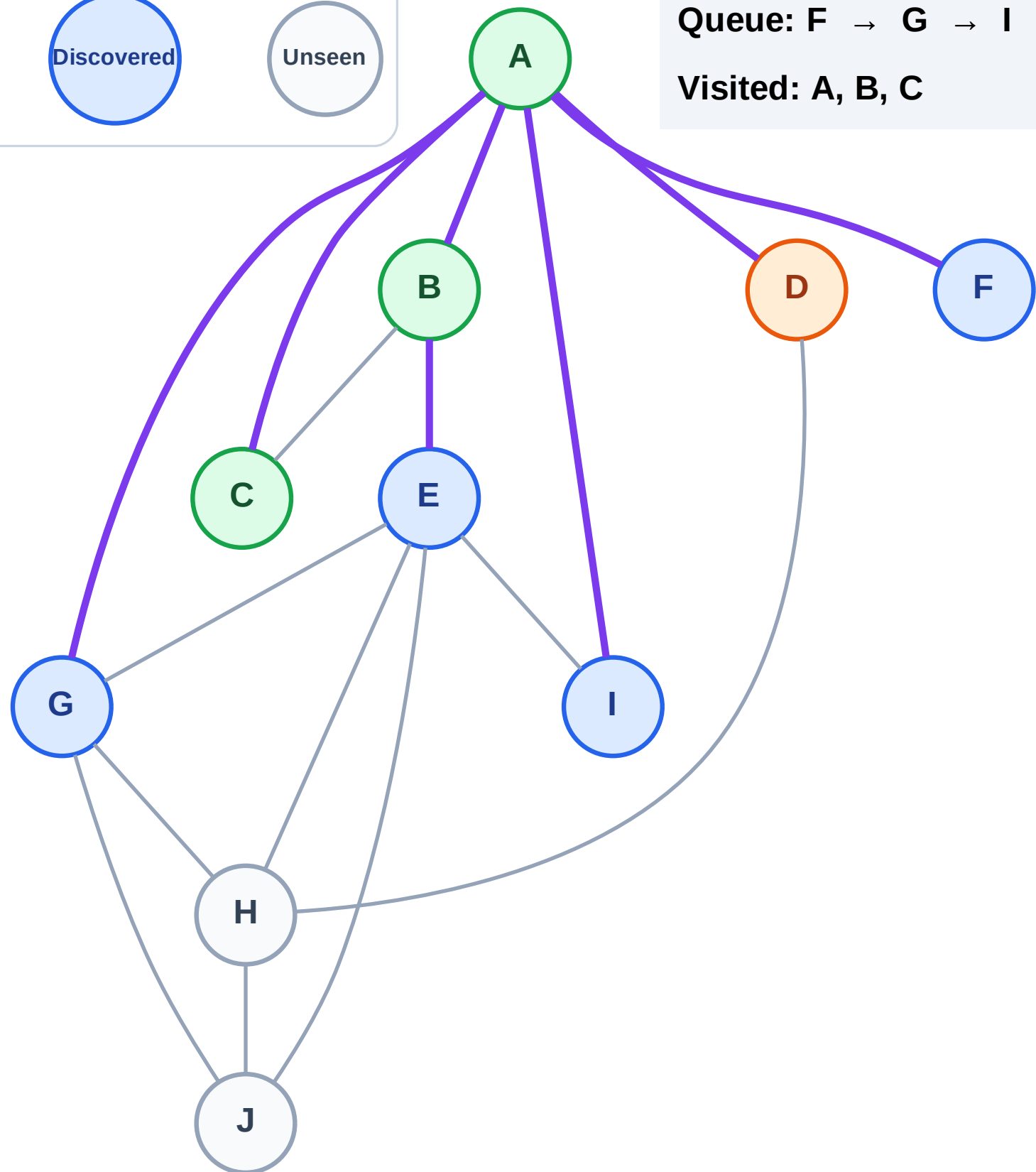
Step 8: Process node D

Legend



Queue: F → G → I → E

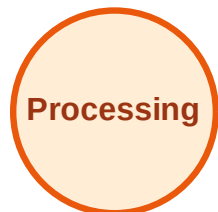
Visited: A, B, C



BFS Traversal

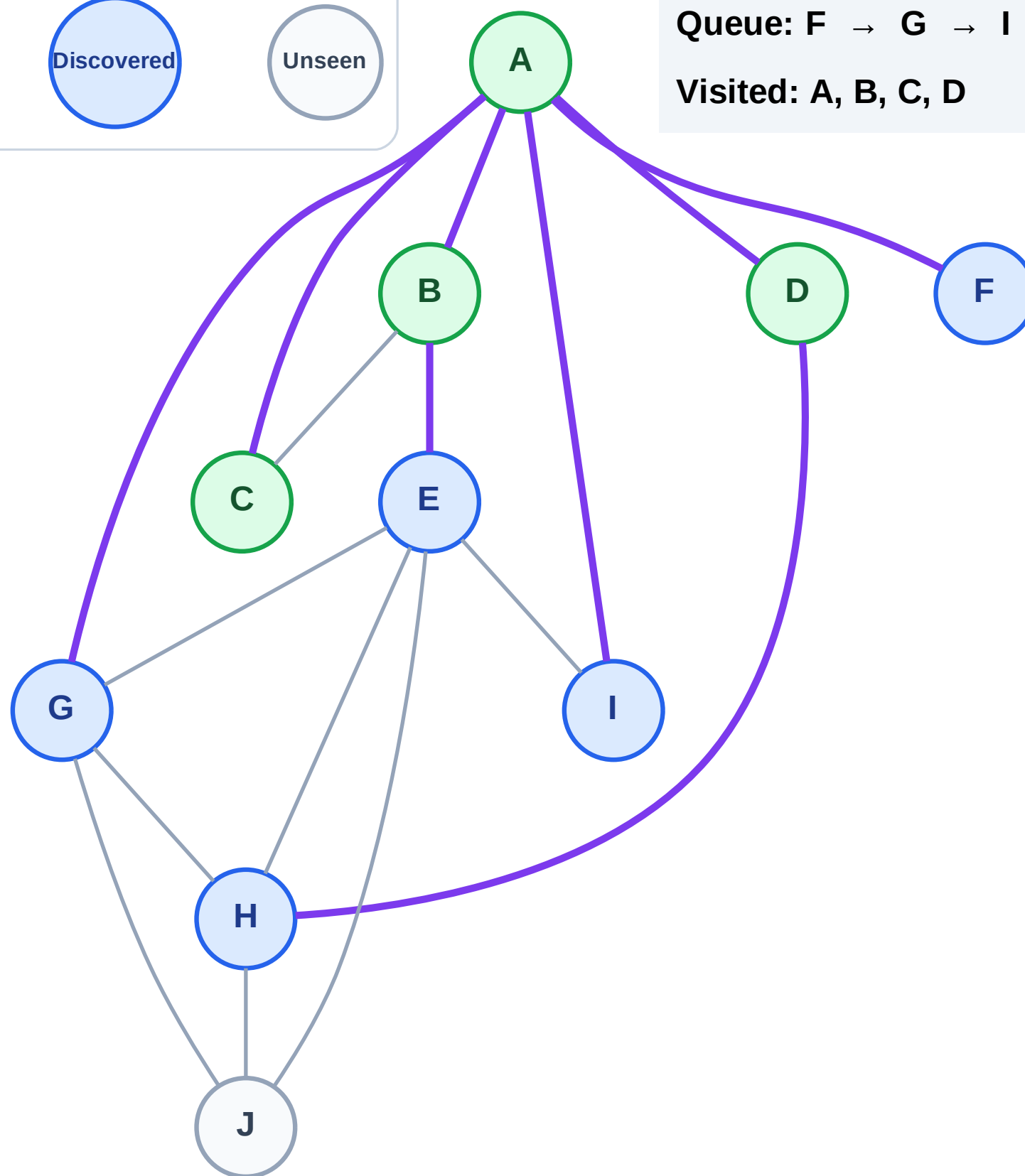
Step 9: Discover H from D

Legend



Queue: F → G → I → E → H

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

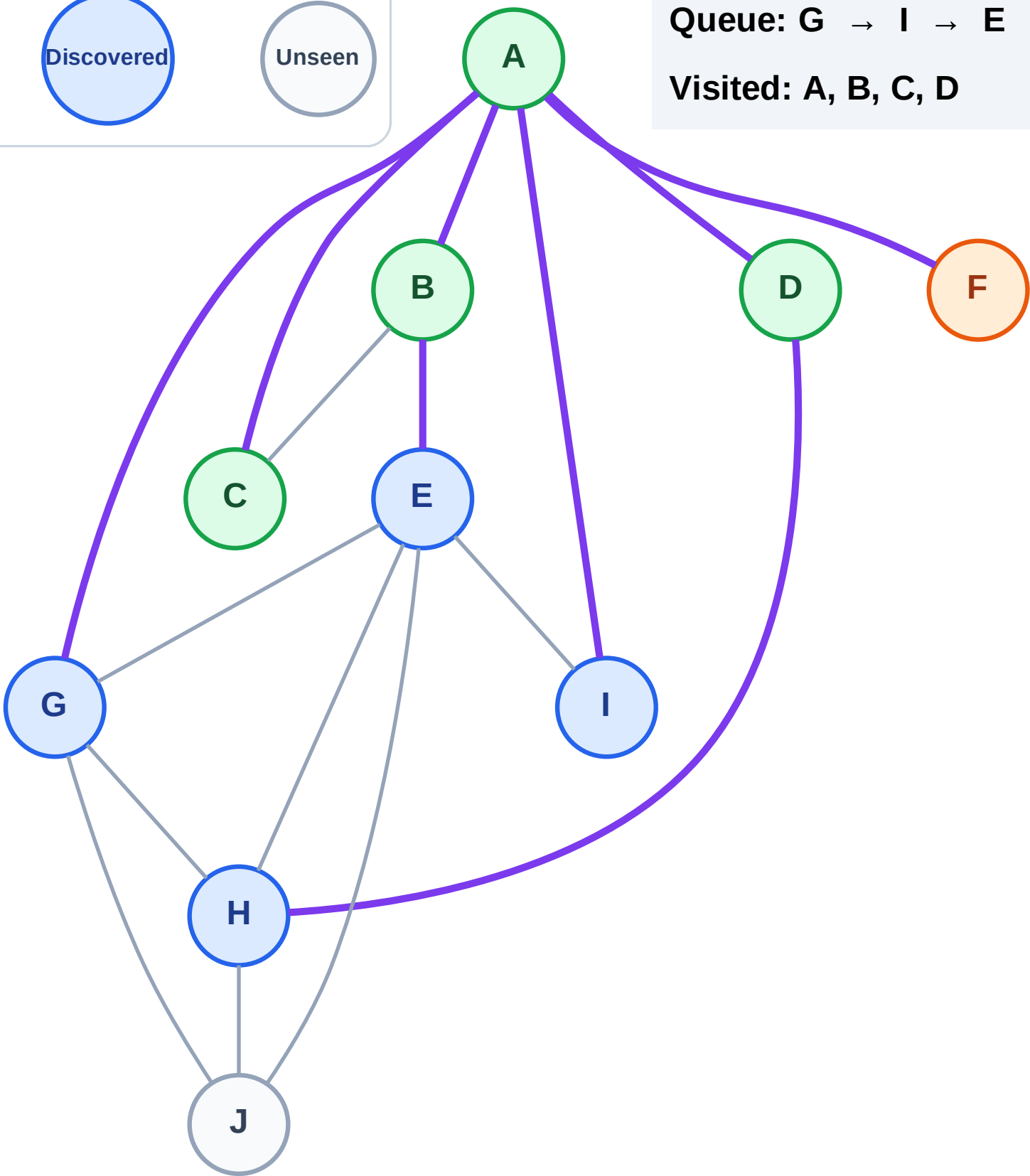
Processing

Discovered

Unseen

Queue: G → I → E → H

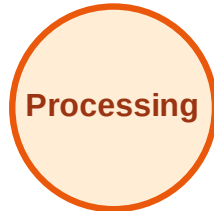
Visited: A, B, C, D



BFS Traversal

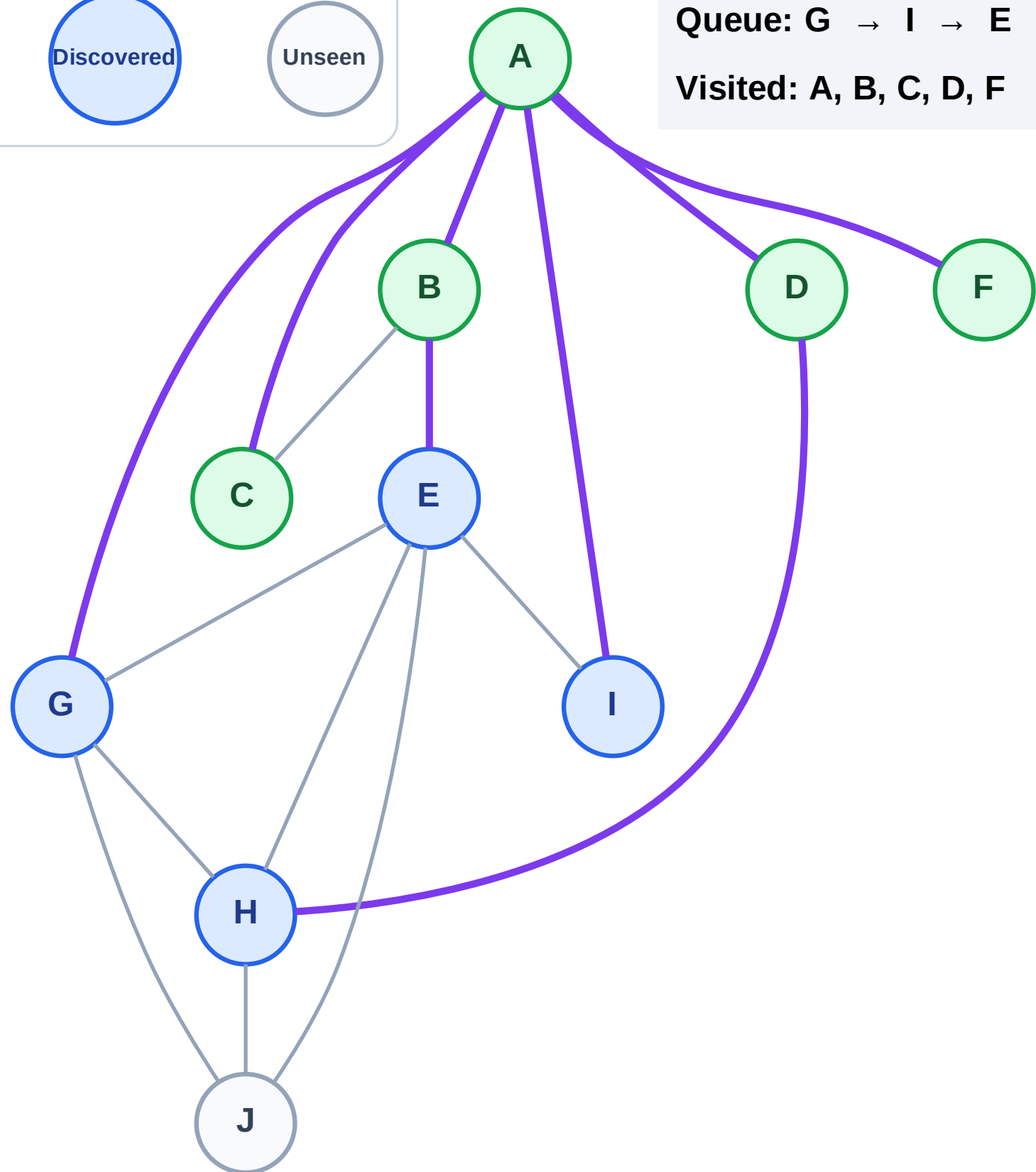
Step 11: No new neighbors from F

Legend



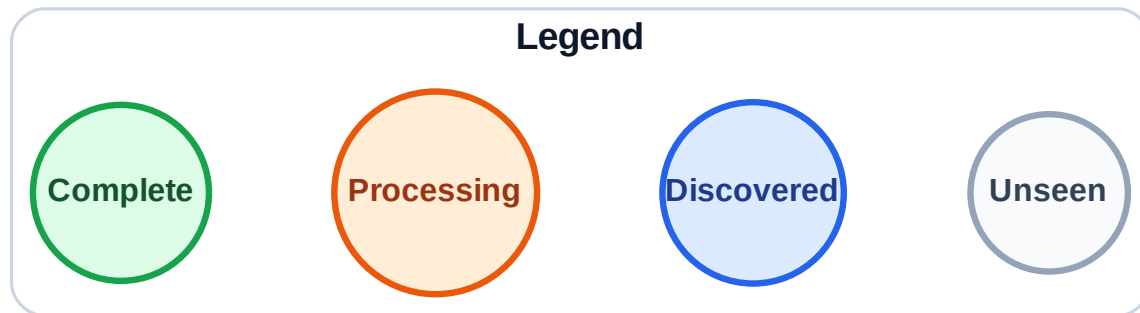
Queue: G → I → E → H

Visited: A, B, C, D, F

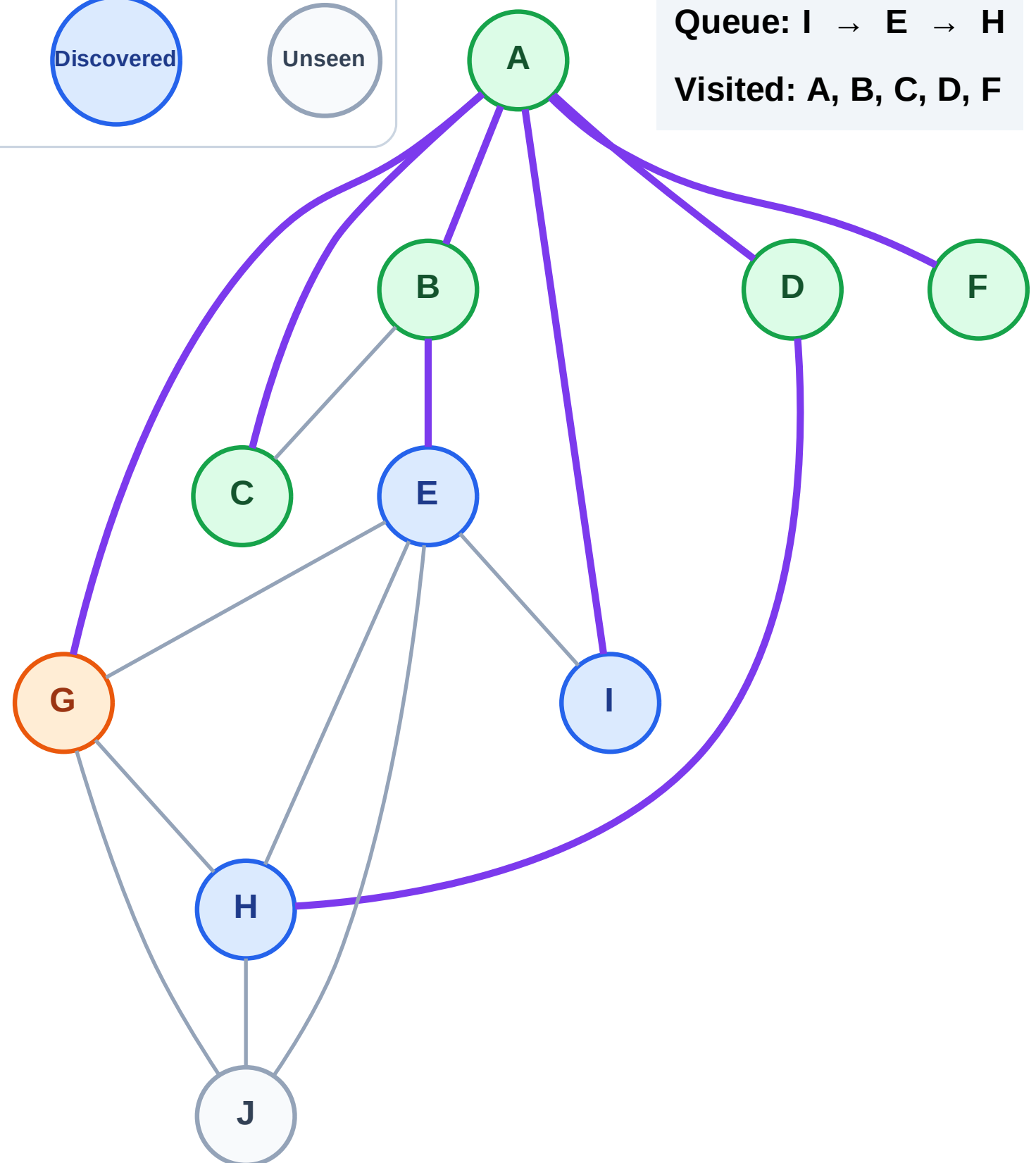


BFS Traversal

Step 12: Process node G



Queue: I → E → H
Visited: A, B, C, D, F



BFS Traversal

Step 13: Discover J from G

Legend

Complete

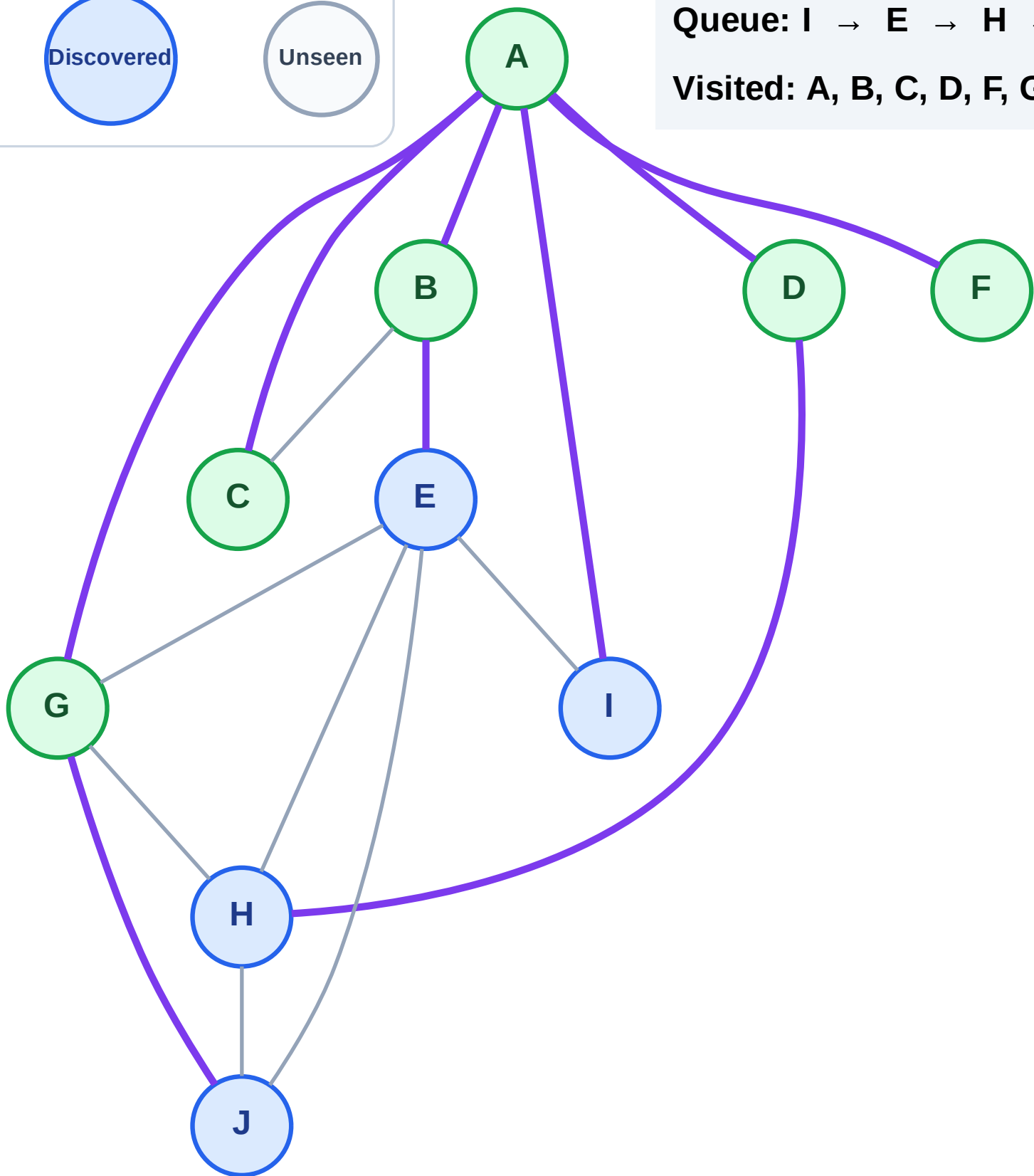
Processing

Discovered

Unseen

Queue: I → E → H → J

Visited: A, B, C, D, F, G



BFS Traversal

Step 14: Process node I

Legend

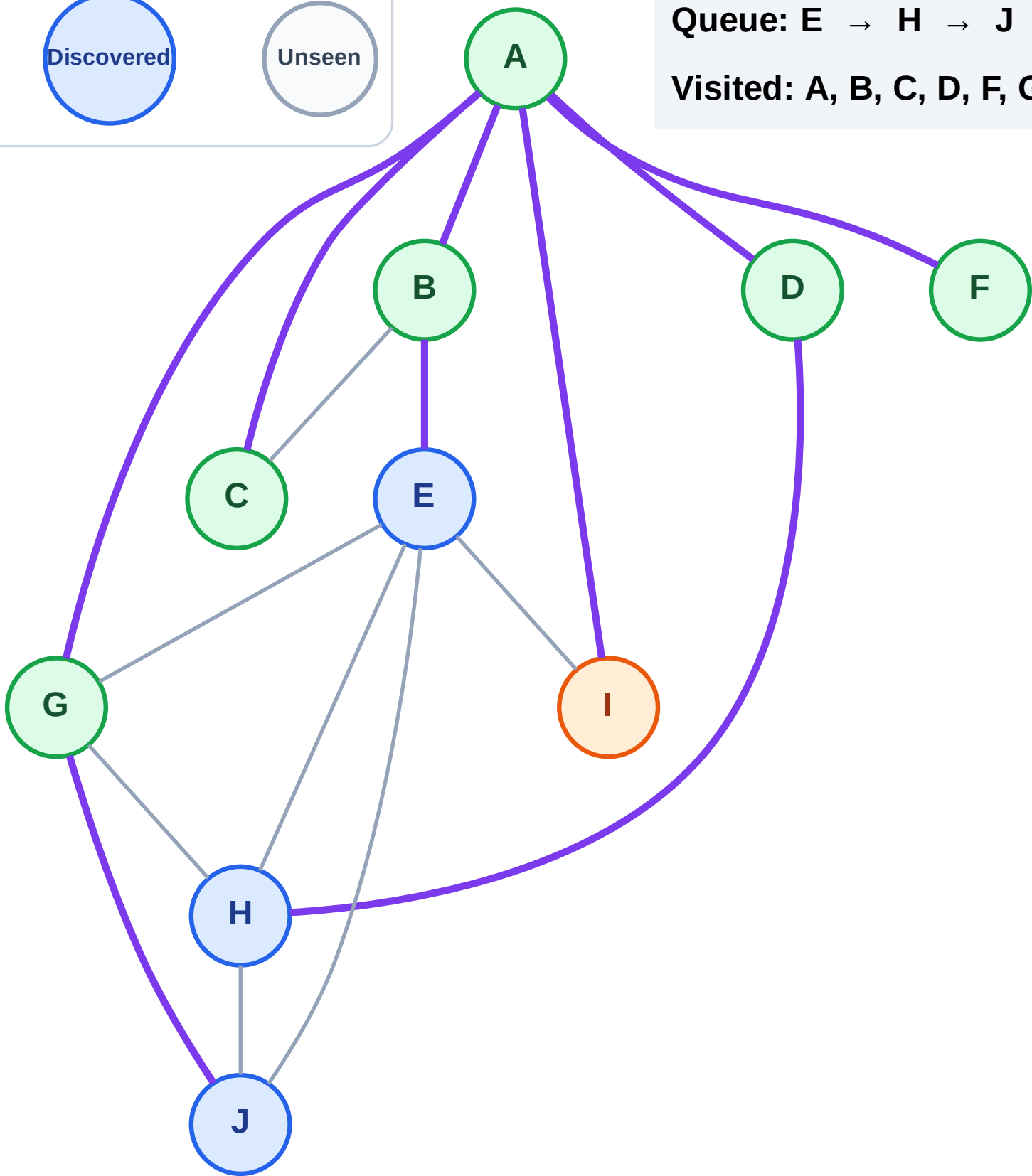
Complete

Processing

Discovered

Unseen

Queue: E → H → J
Visited: A, B, C, D, F, G



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

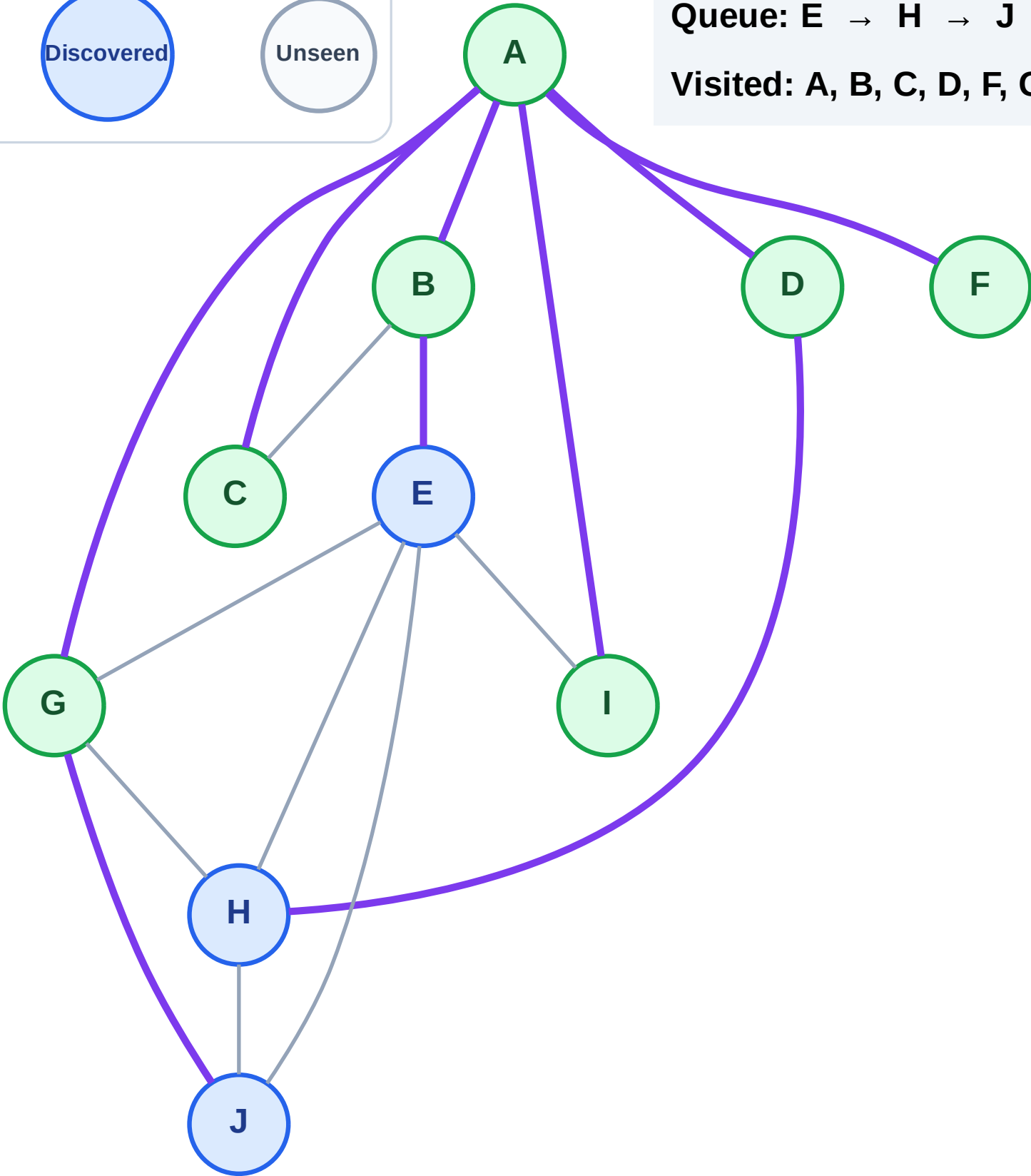
Processing

Discovered

Unseen

Queue: E → H → J

Visited: A, B, C, D, F, G, I



BFS Traversal

Step 16: Process node E

Legend

Complete

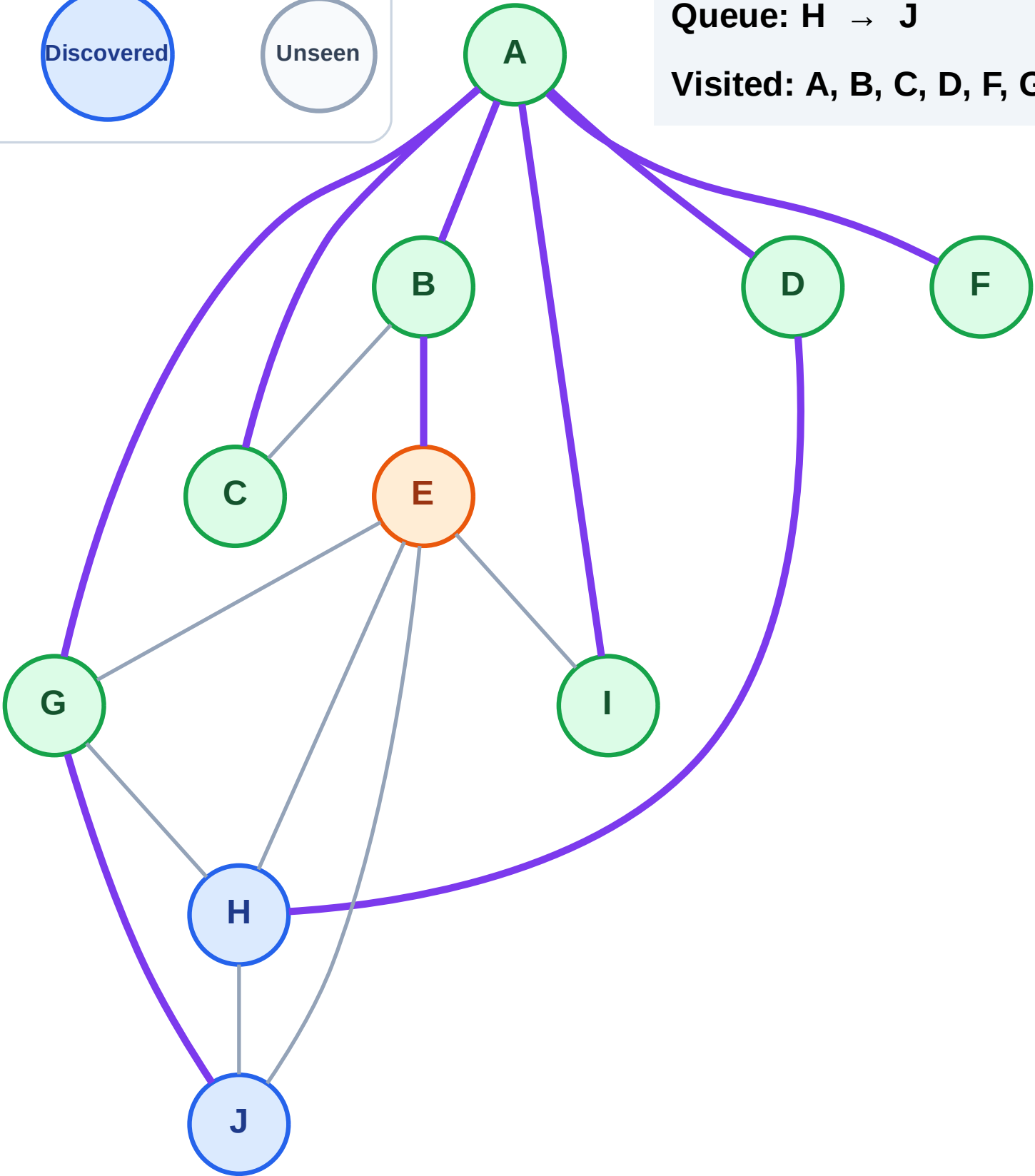
Processing

Discovered

Unseen

Queue: H → J

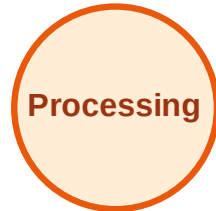
Visited: A, B, C, D, F, G, I



BFS Traversal

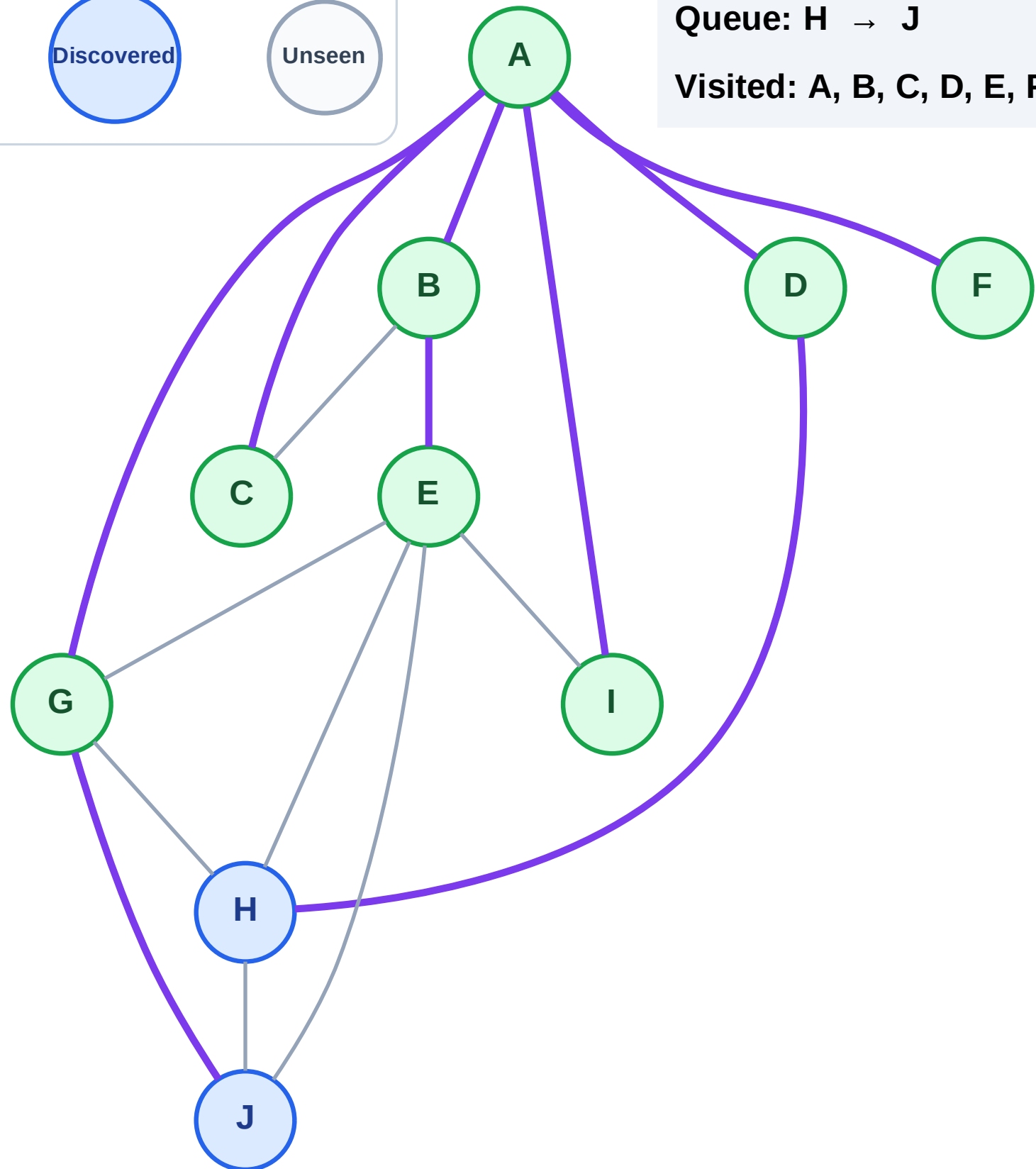
Step 17: No new neighbors from E

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

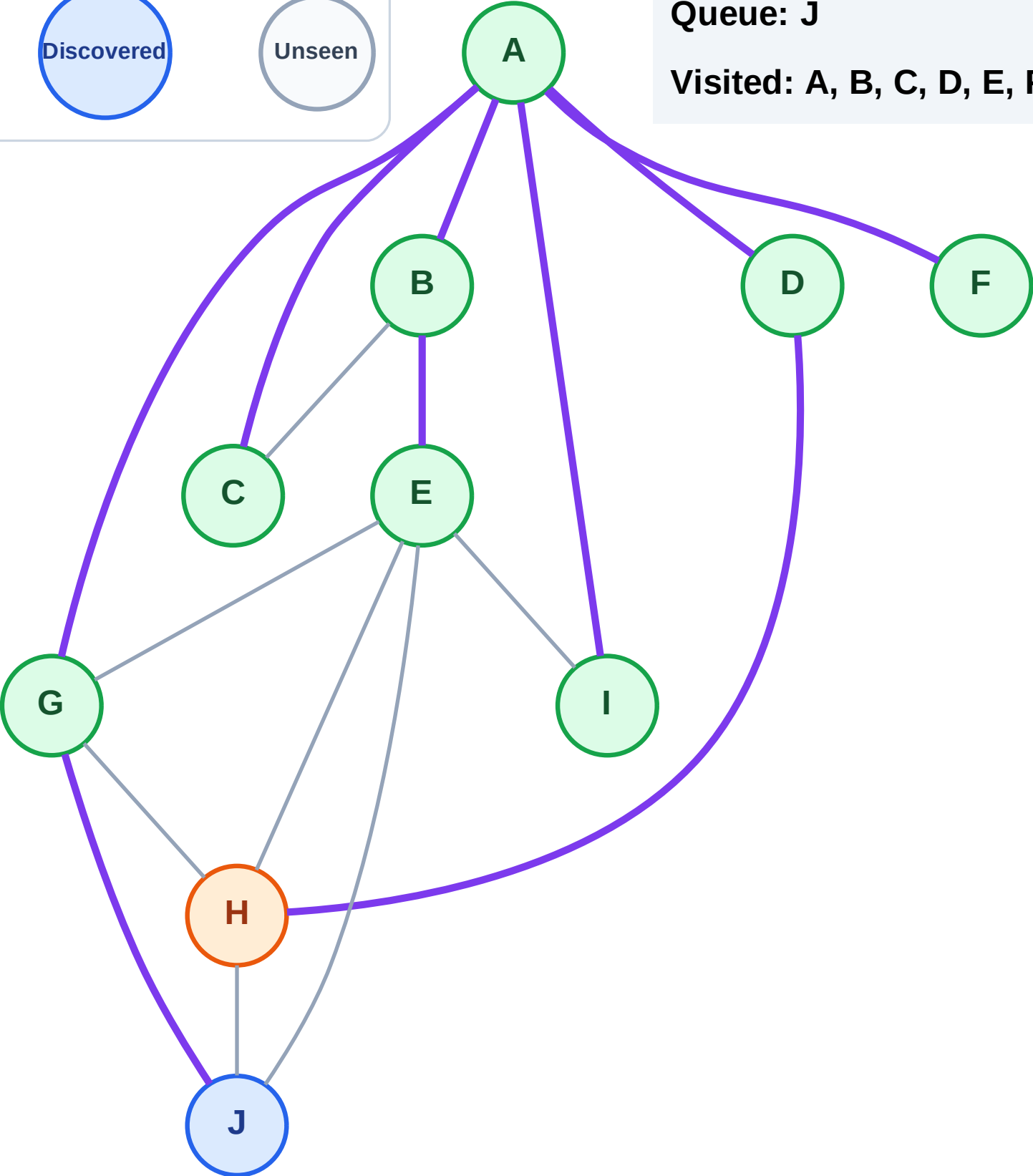
Processing

Discovered

Unseen

Queue: J

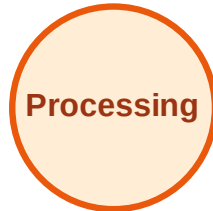
Visited: A, B, C, D, E, F, G, I



BFS Traversal

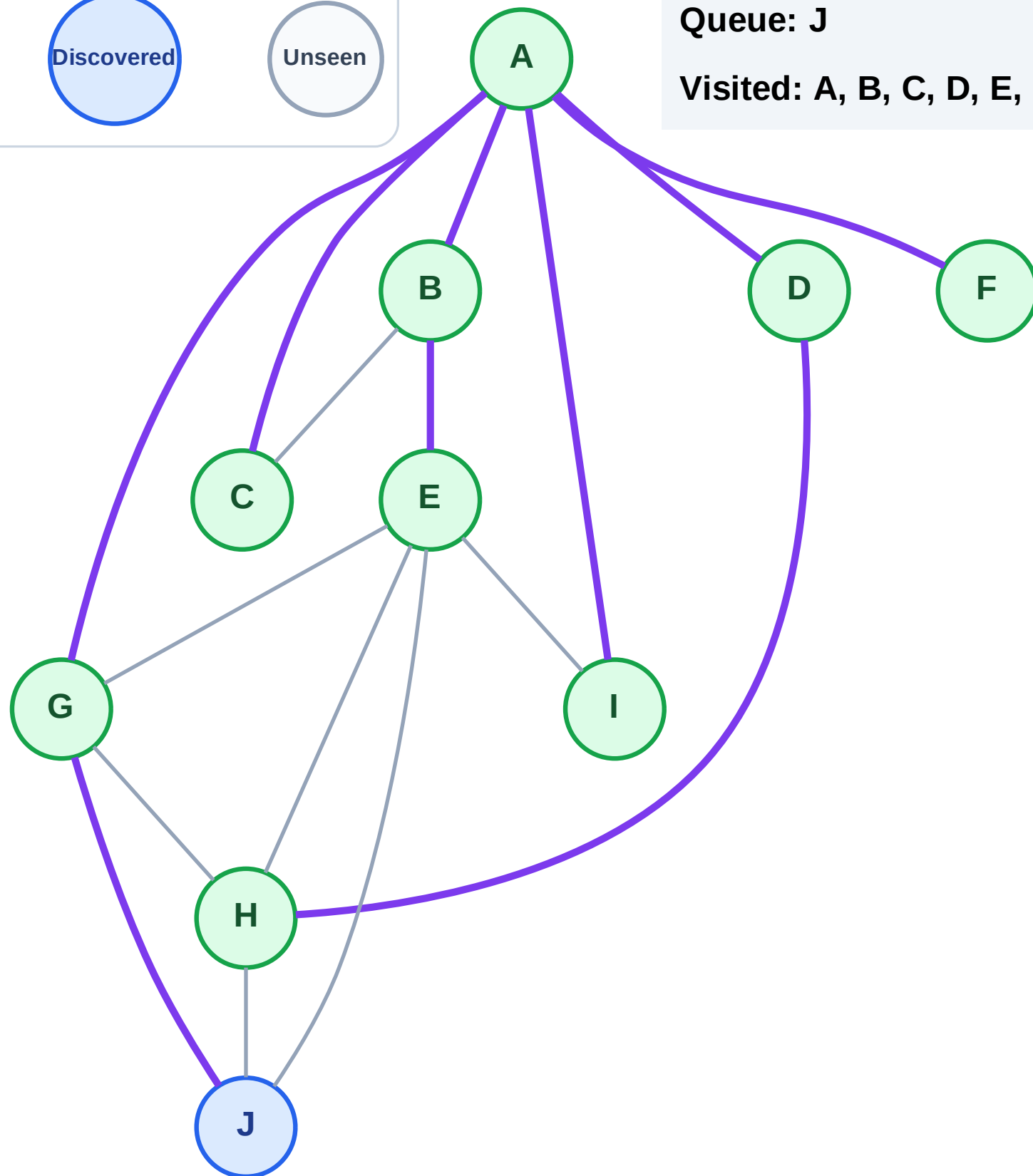
Step 19: No new neighbors from H

Legend



Queue: J

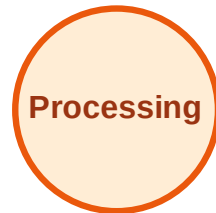
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

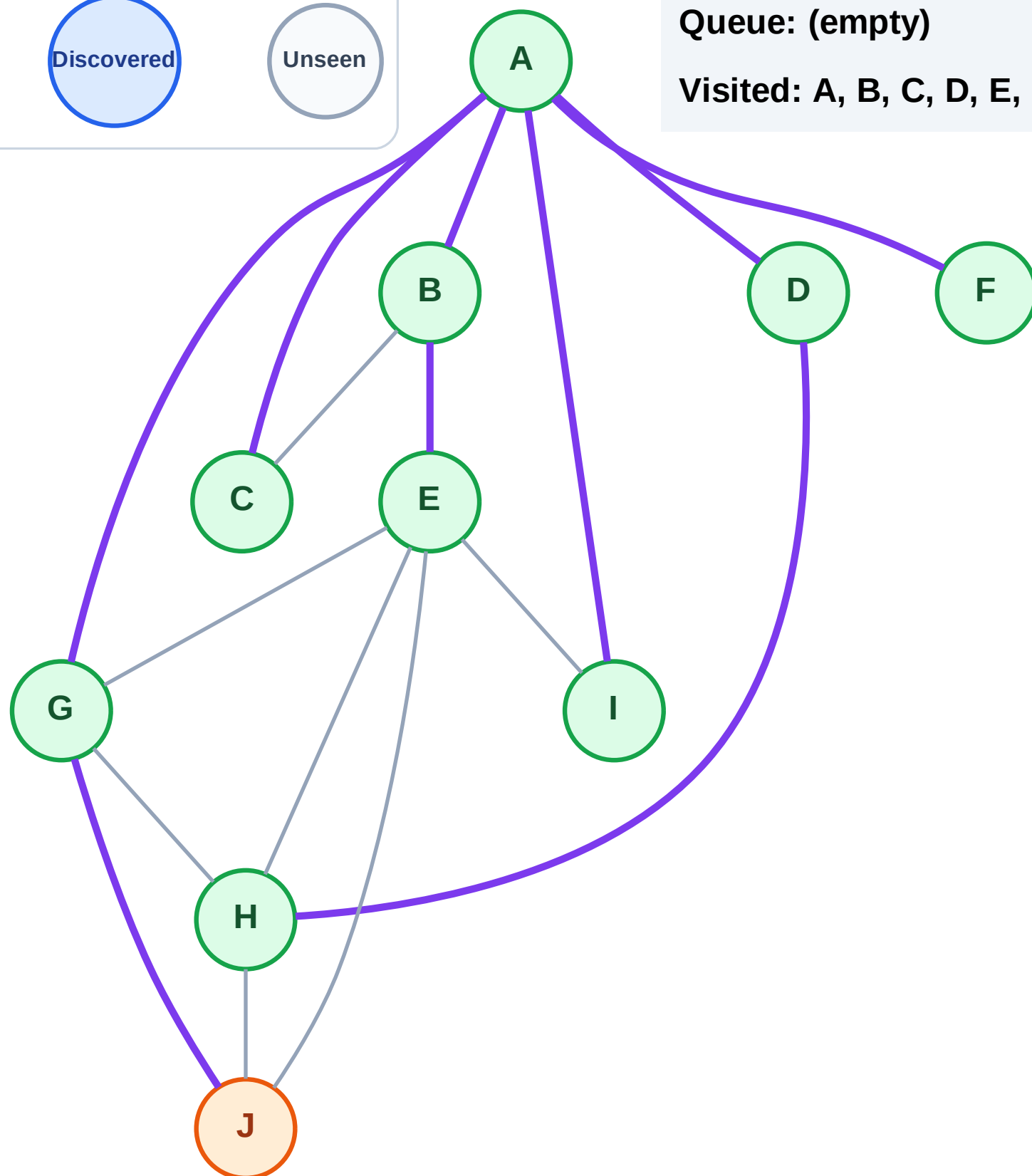
Step 20: Process node J

Legend



Queue: (empty)

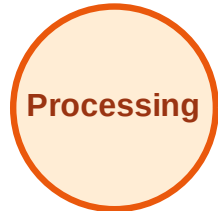
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

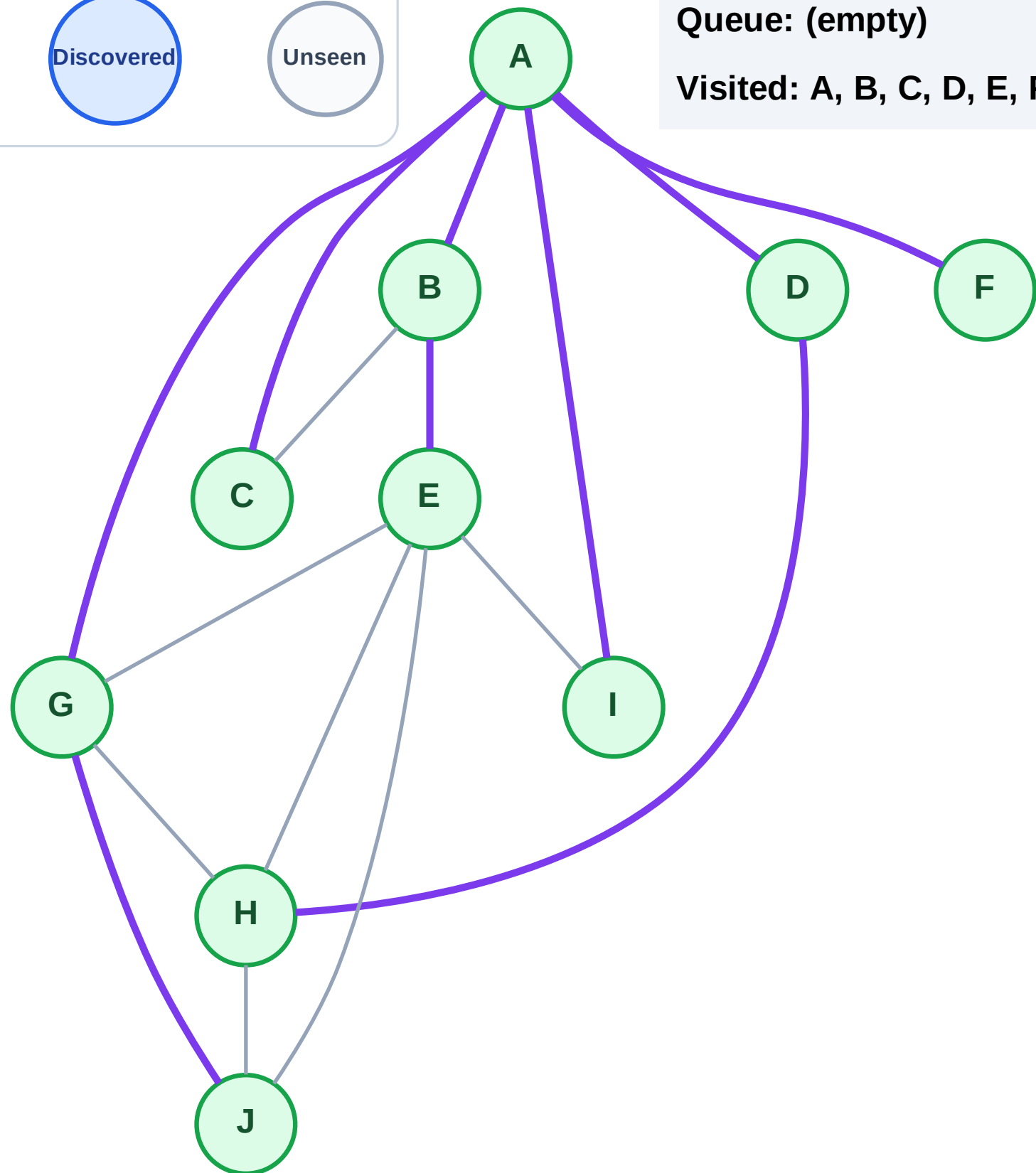
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

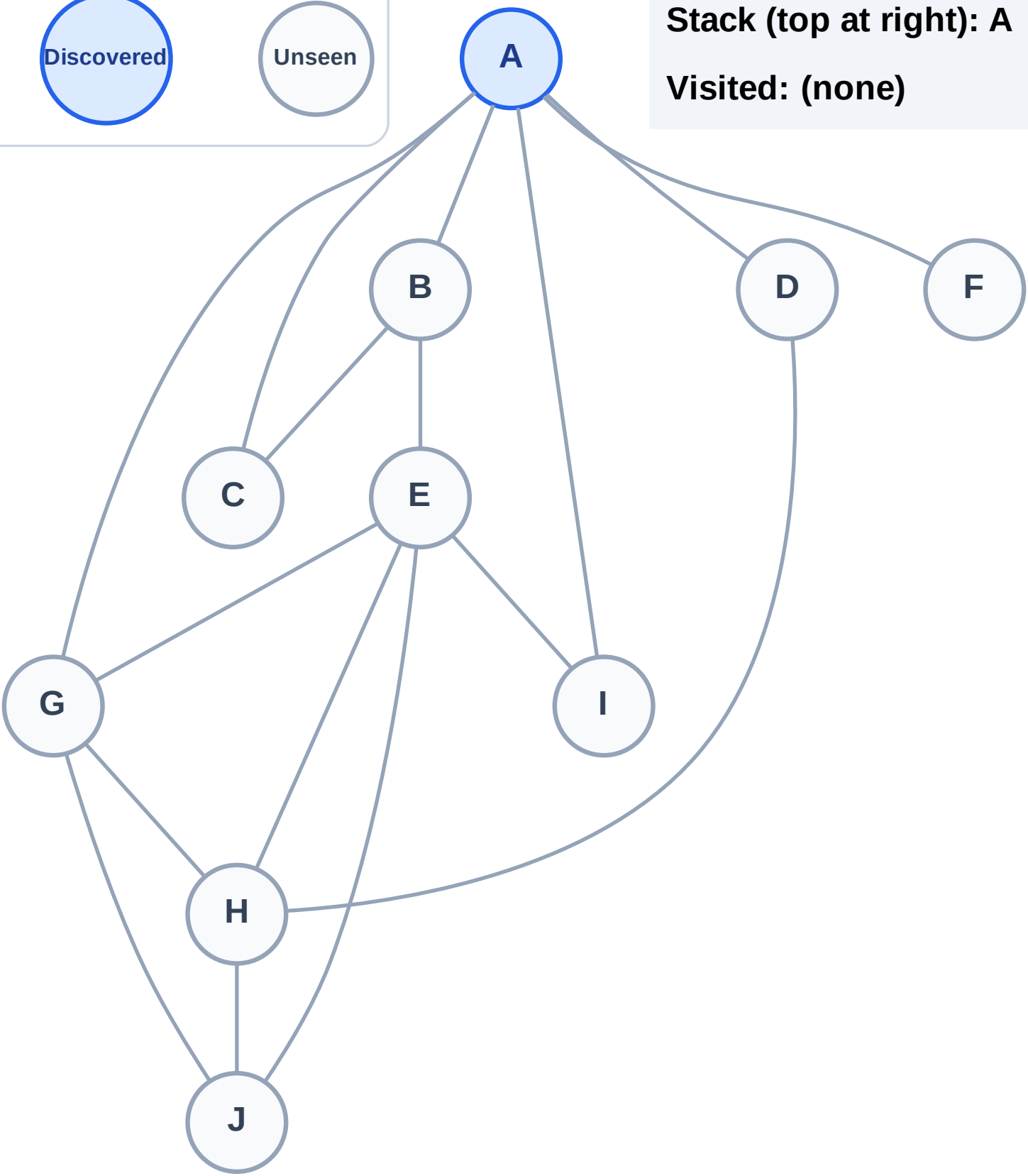
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

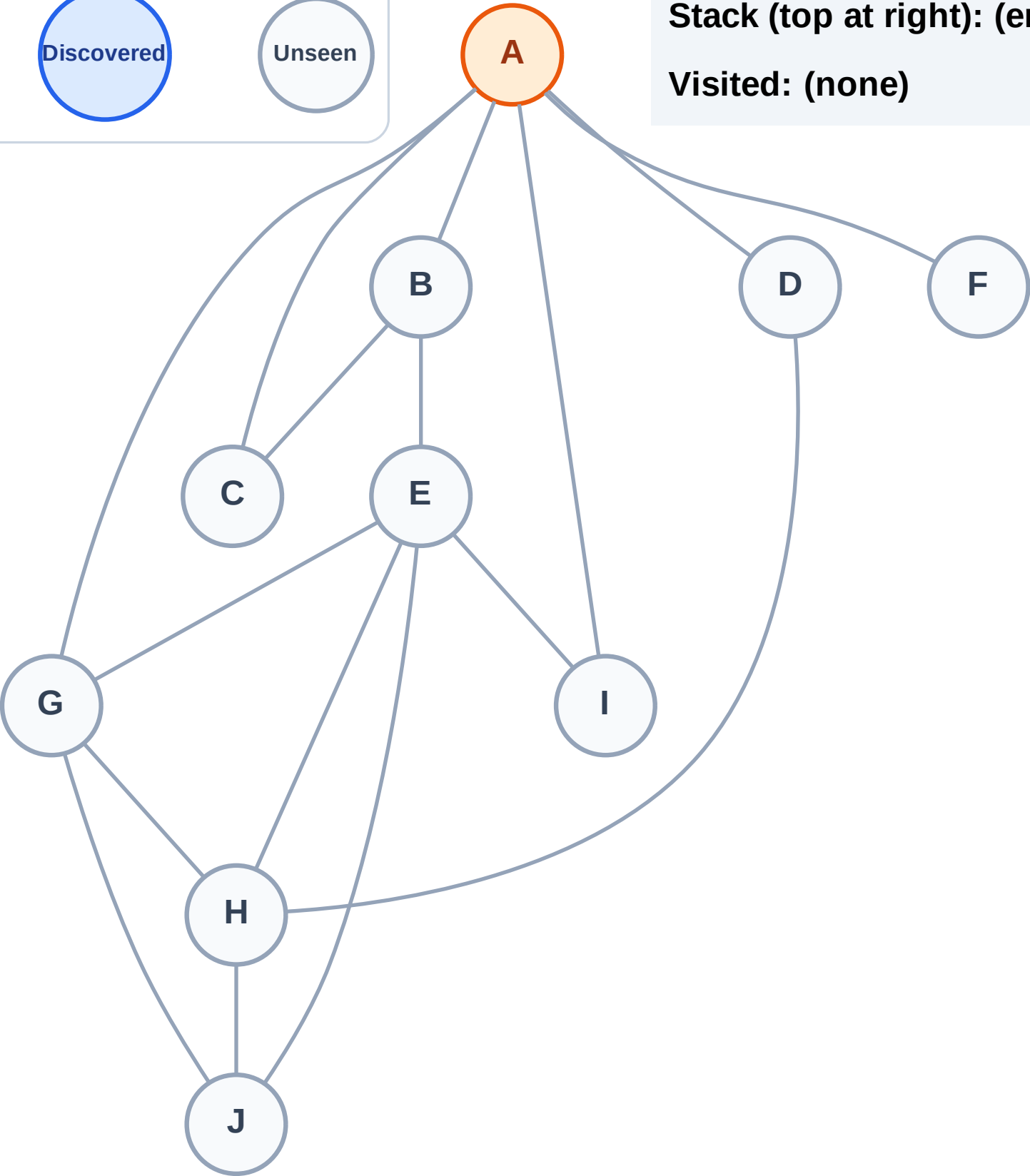
Complete

Processing

Discovered

Unseen

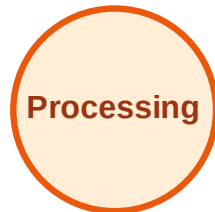
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

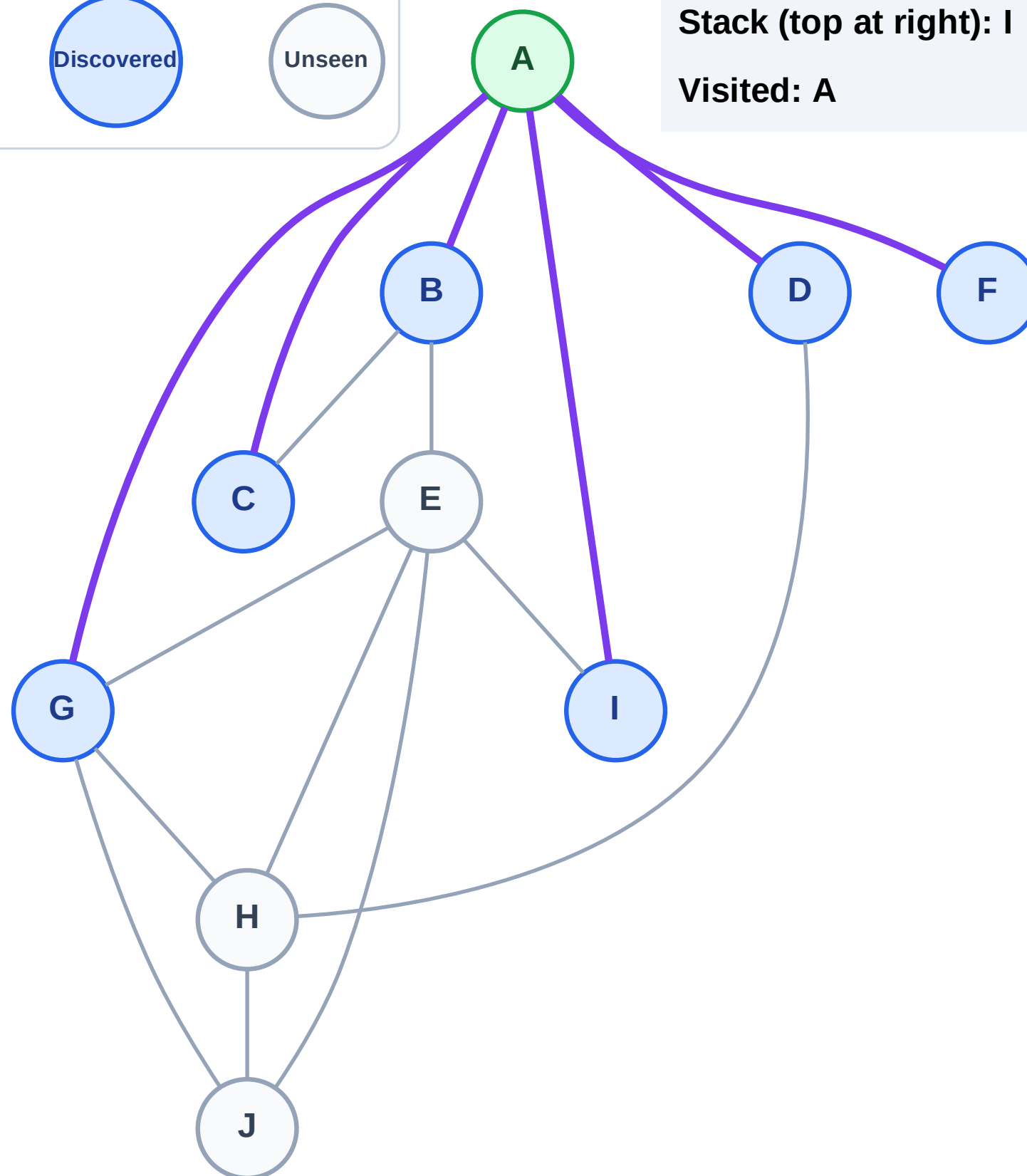
Step 3: Discover I, G, F, D, C, B from A

Legend



Stack (top at right): I | G | F | D | C | B

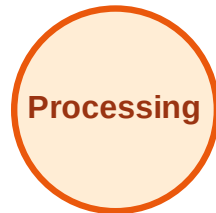
Visited: A



DFS Traversal

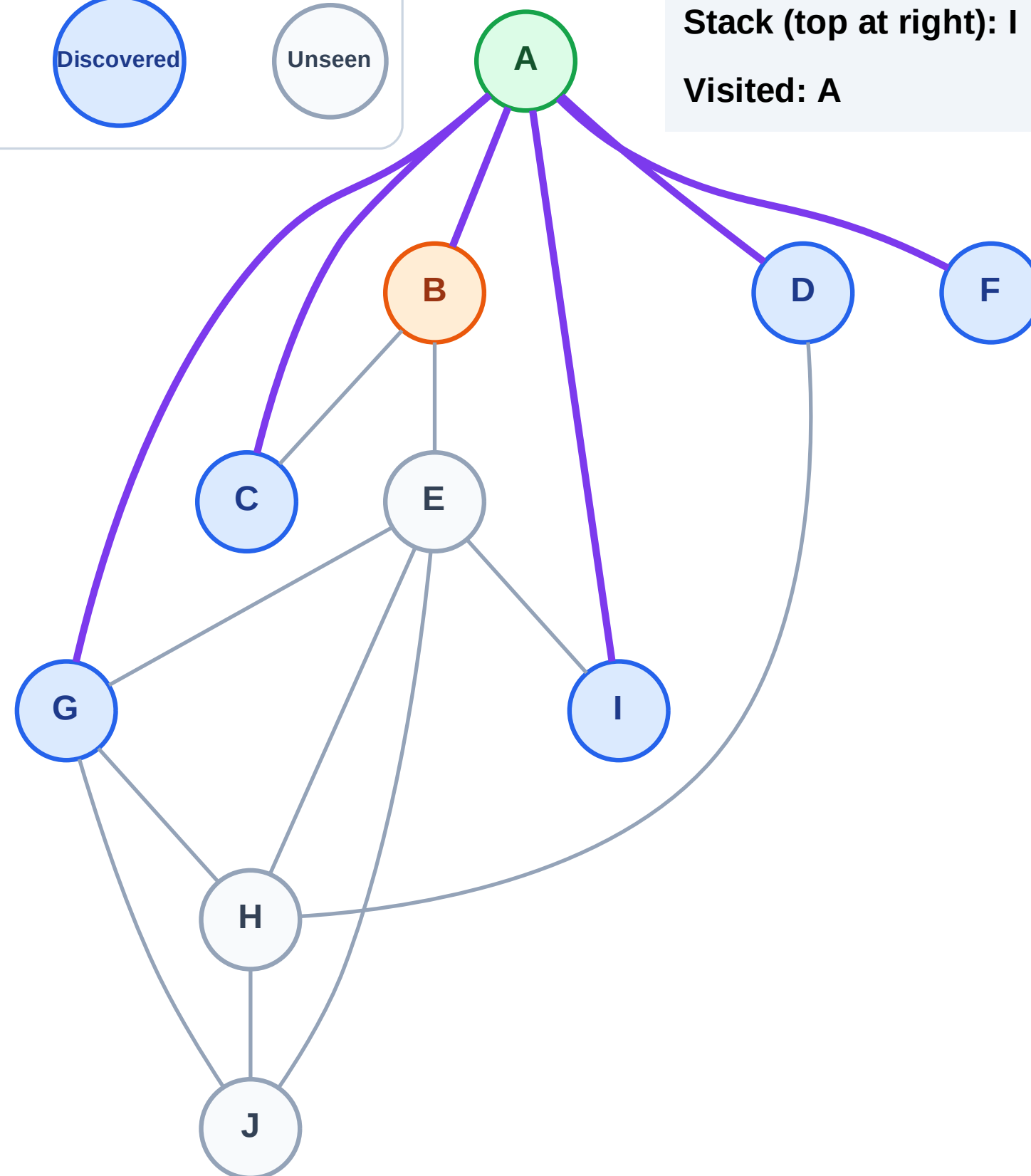
Step 4: Process node B

Legend



Stack (top at right): I | G | F | D | C

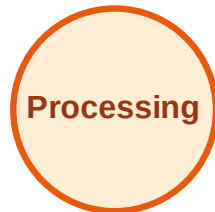
Visited: A



DFS Traversal

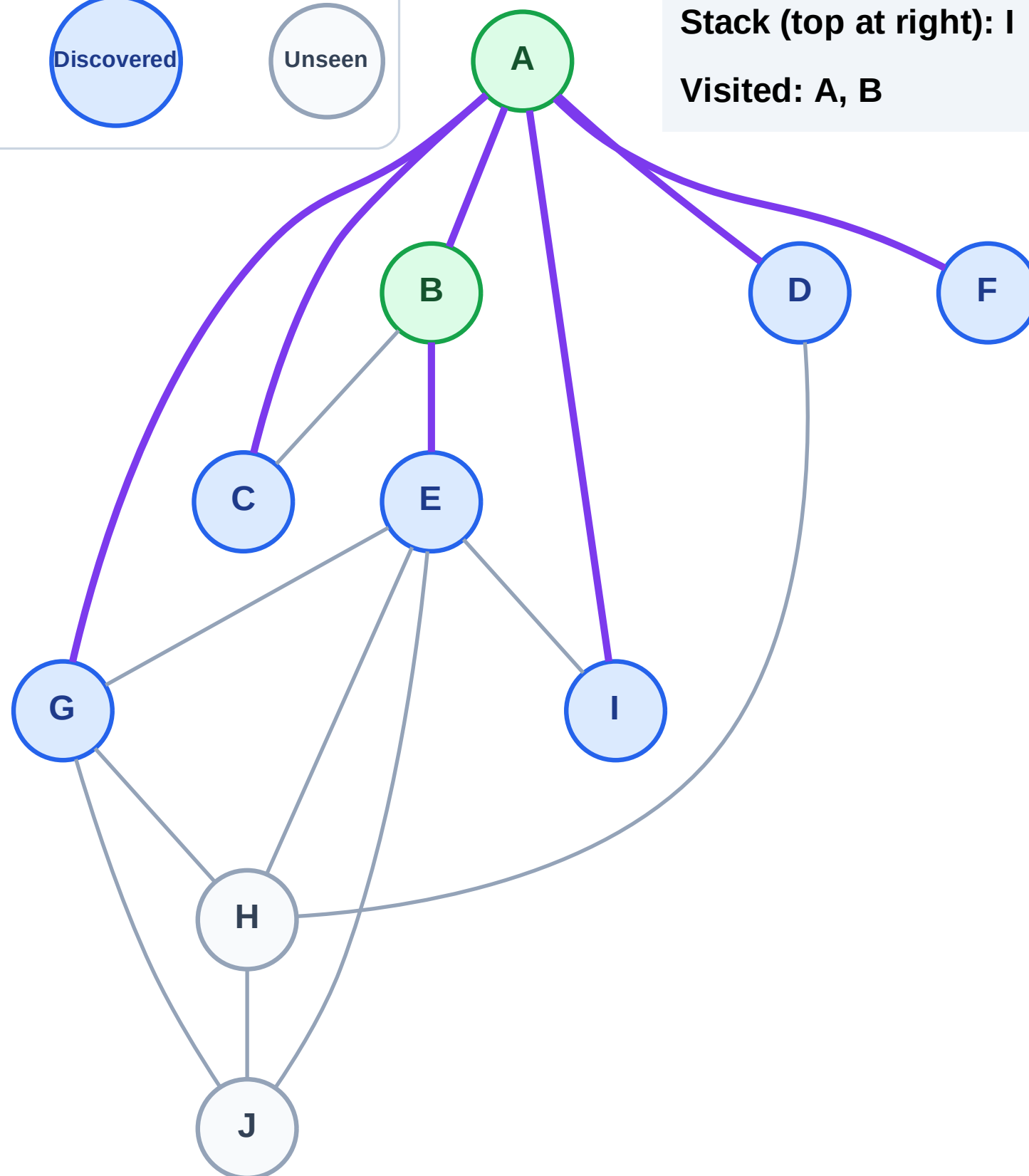
Step 5: Discover E from B

Legend



Stack (top at right): I | G | F | D | C | E

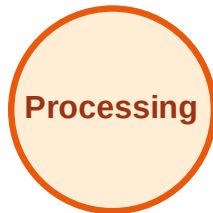
Visited: A, B



DFS Traversal

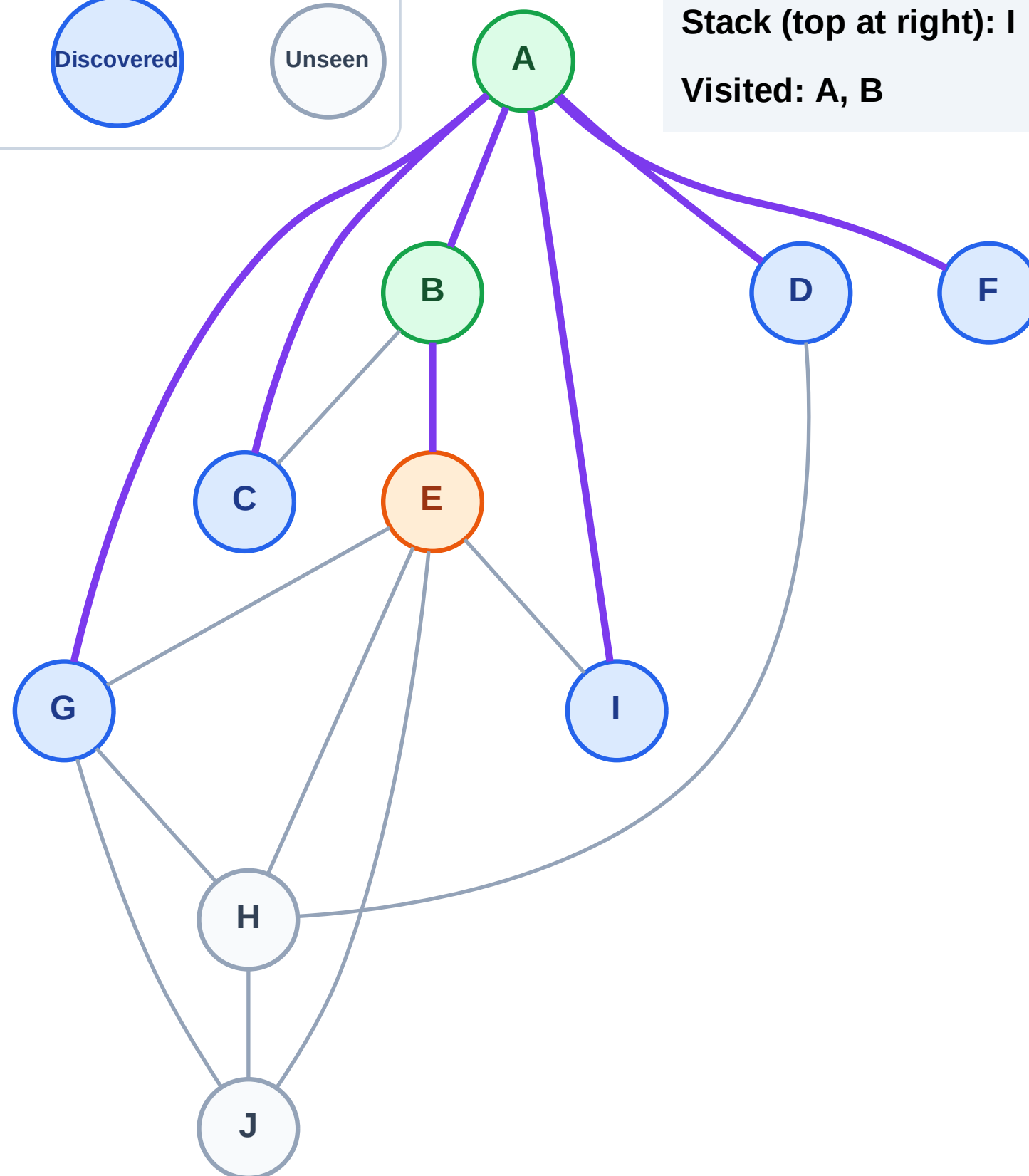
Step 6: Process node E

Legend



Stack (top at right): I | G | F | D | C

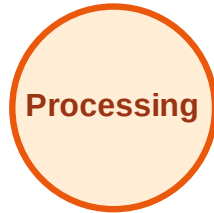
Visited: A, B



DFS Traversal

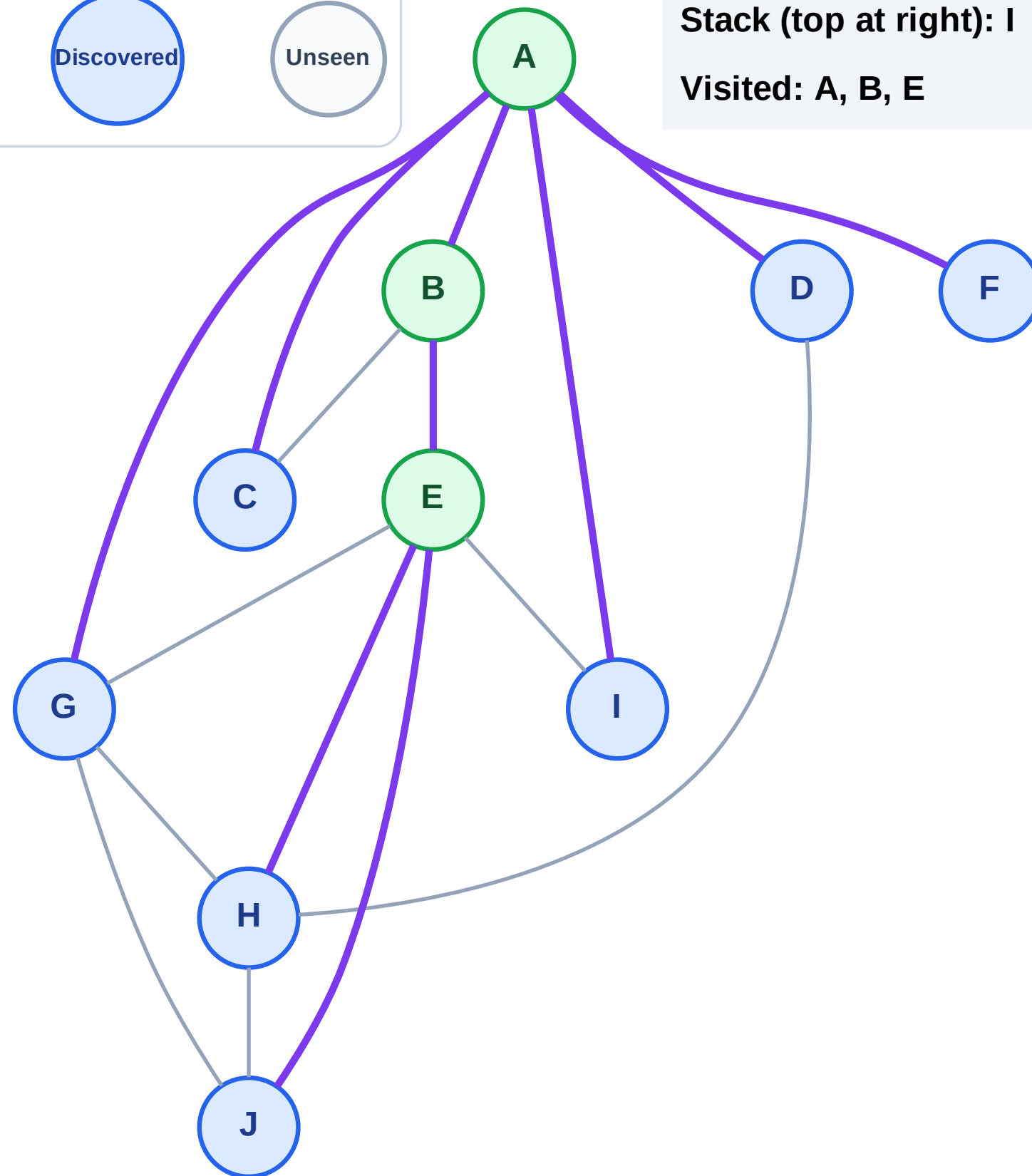
Step 7: Discover J, H from E

Legend



Stack (top at right): I | G | F | D | C | J | H

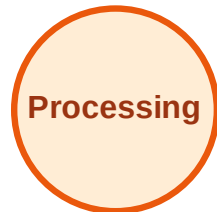
Visited: A, B, E



DFS Traversal

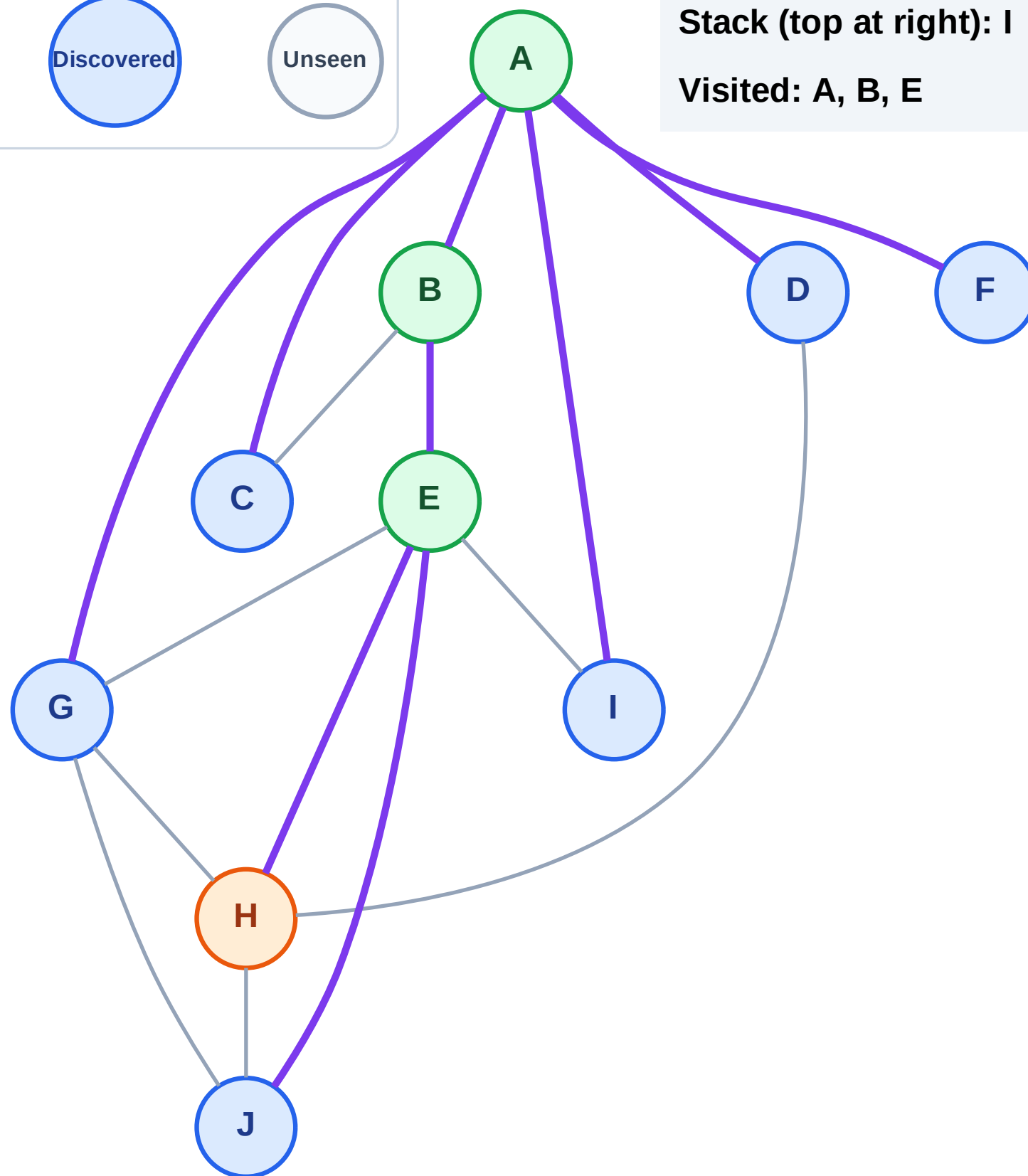
Step 8: Process node H

Legend



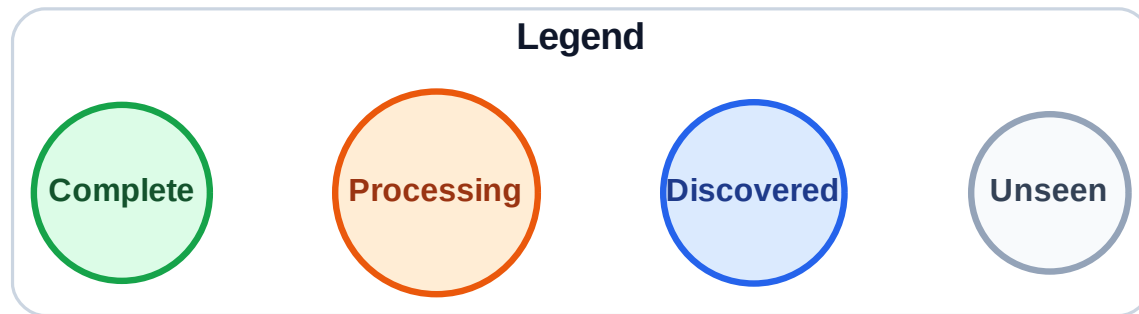
Stack (top at right): I | G | F | D | C | J

Visited: A, B, E



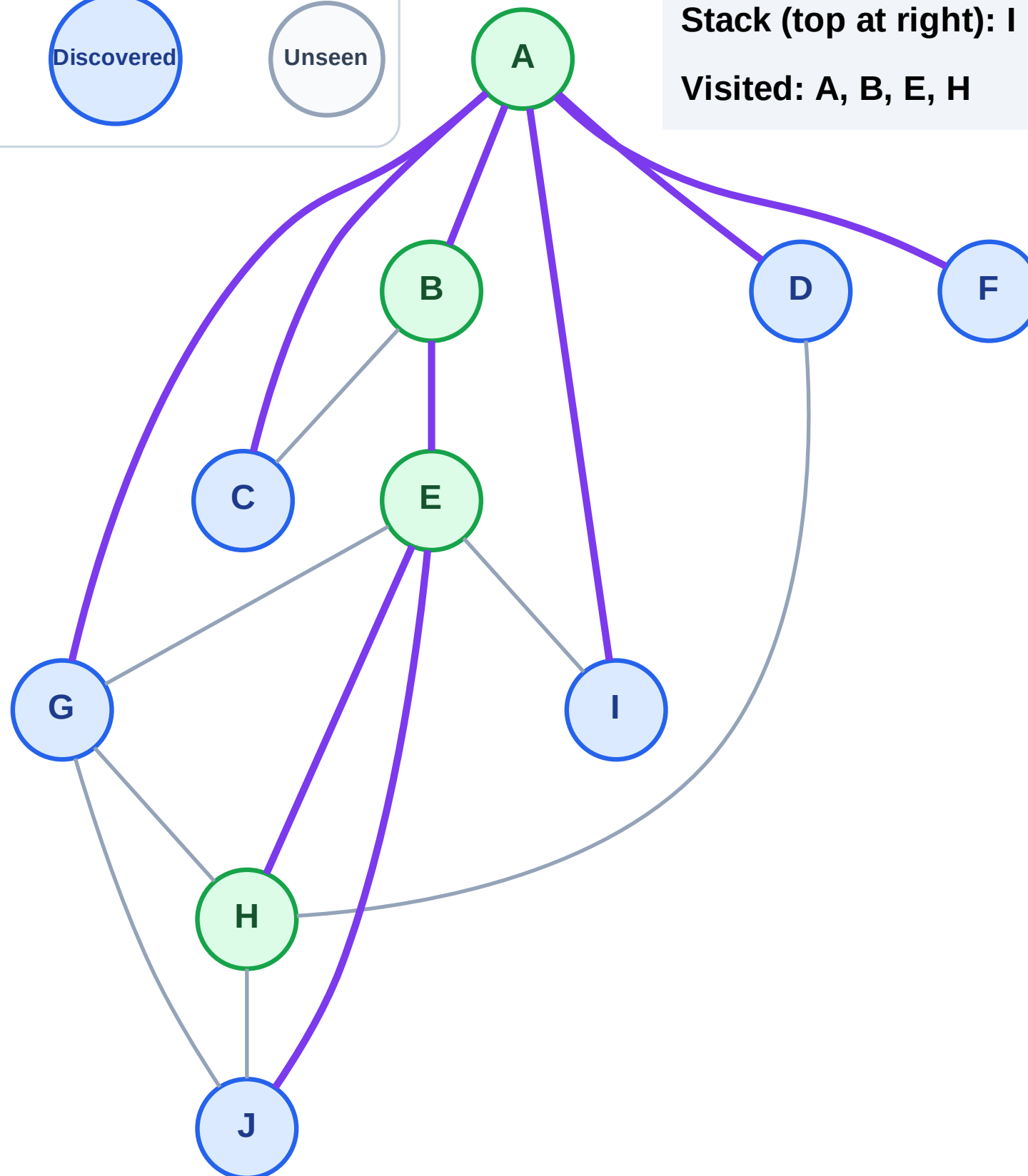
Step 9: No new neighbors from H

Step 9: No new neighbors from H



Stack (top at right): I | G | F | D | C | J

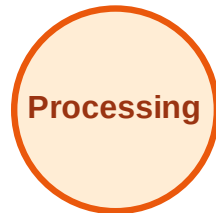
Visited: A, B, E, H



DFS Traversal

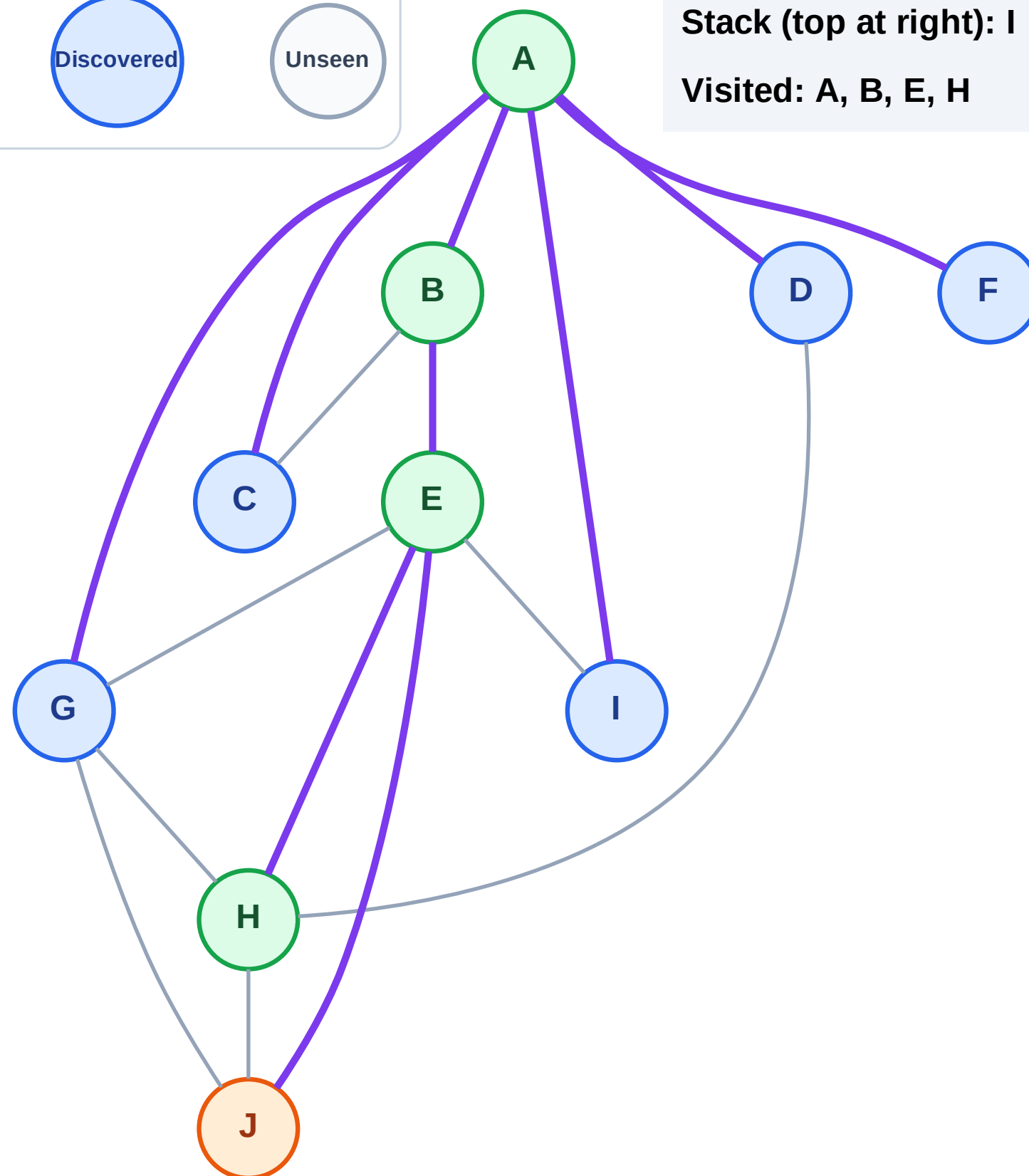
Step 10: Process node J

Legend



Stack (top at right): I | G | F | D | C

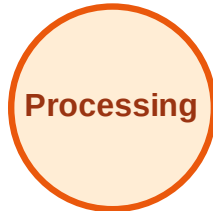
Visited: A, B, E, H



DFS Traversal

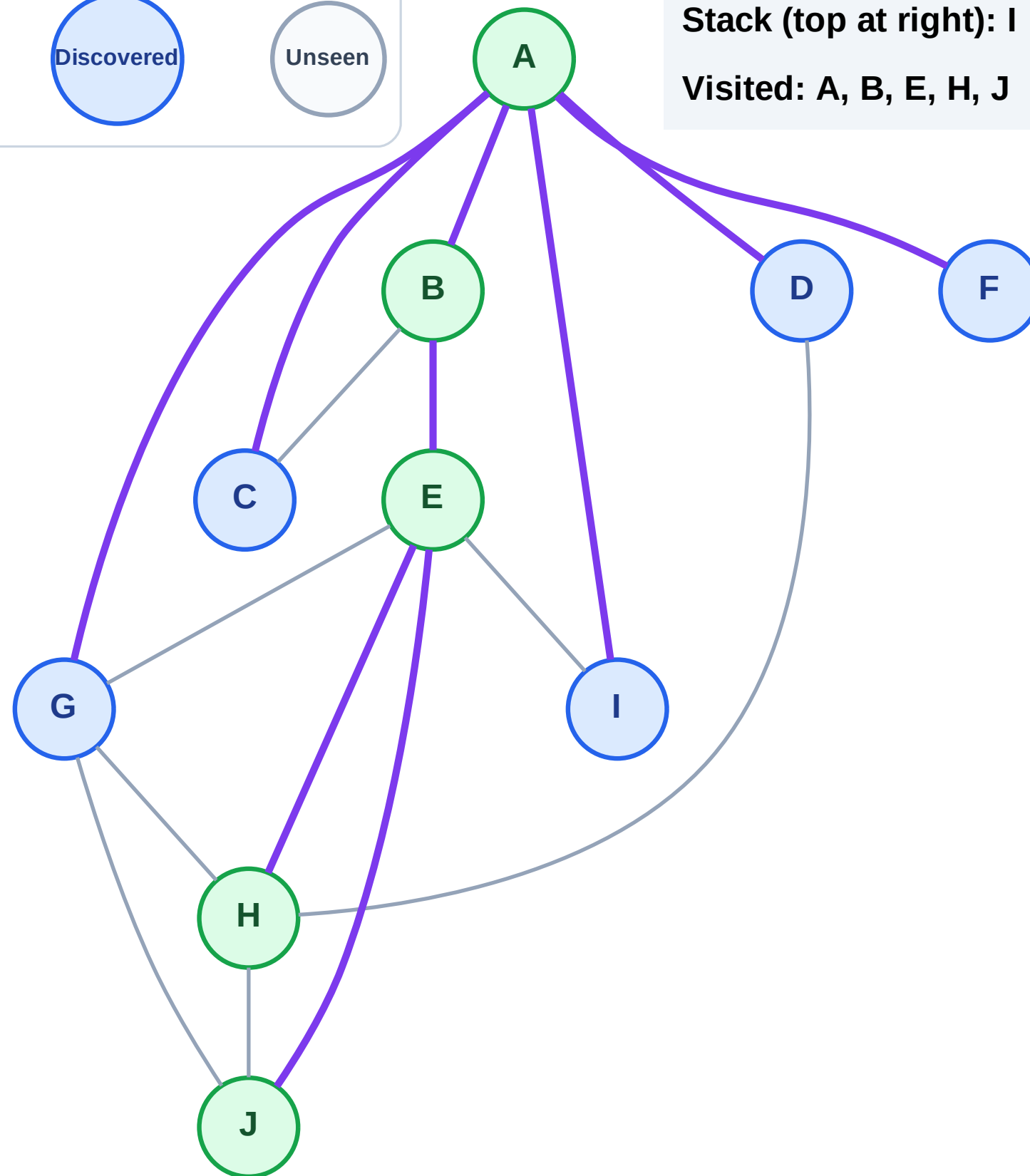
Step 11: No new neighbors from J

Legend



Stack (top at right): I | G | F | D | C

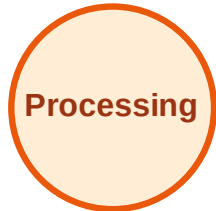
Visited: A, B, E, H, J



DFS Traversal

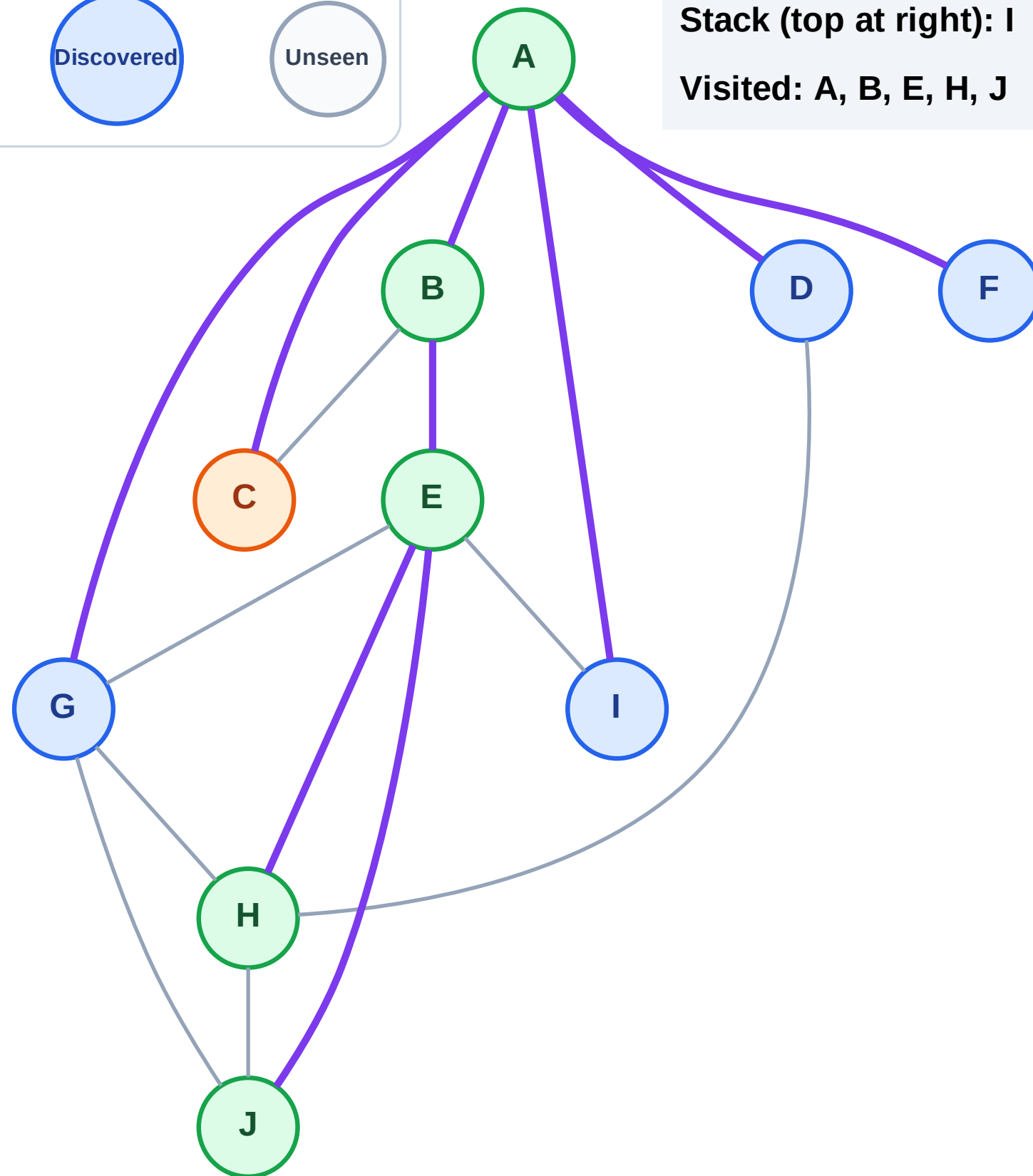
Step 12: Process node C

Legend



Stack (top at right): I | G | F | D

Visited: A, B, E, H, J



DFS Traversal

Step 13: No new neighbors from C

Legend

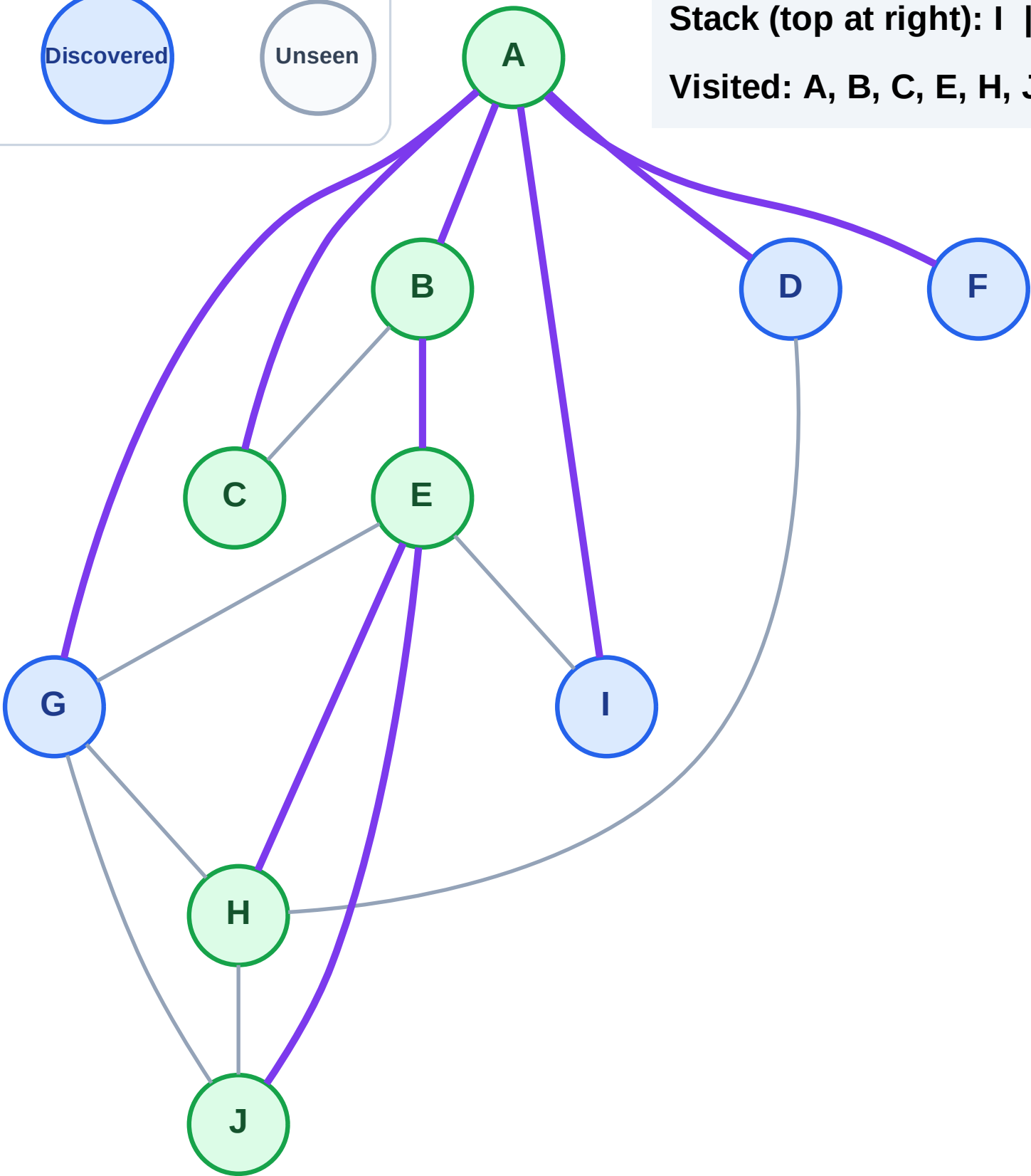
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | F | D
Visited: A, B, C, E, H, J



DFS Traversal

Step 14: Process node D

Legend

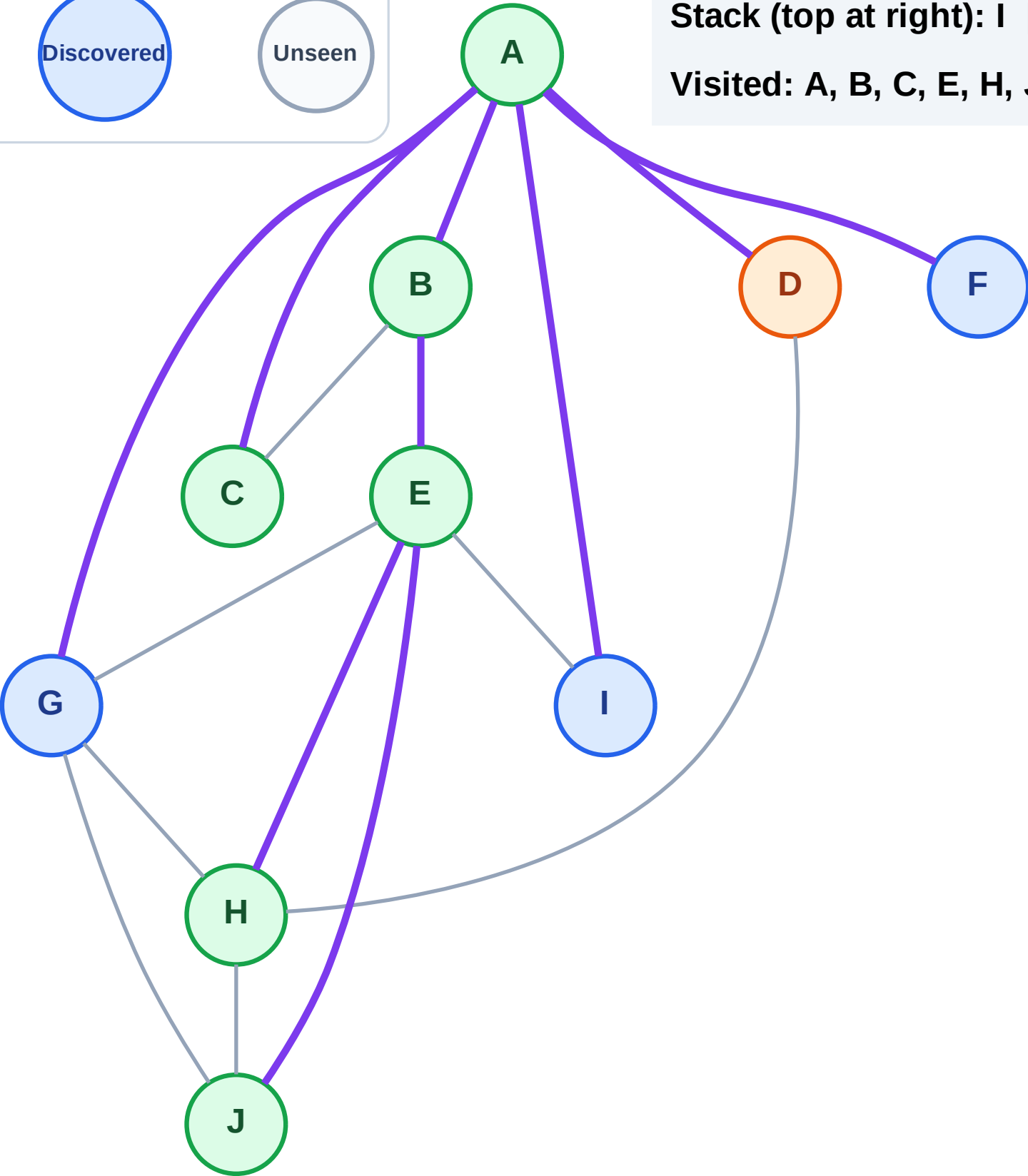
Complete

Processing

Discovered

Unseen

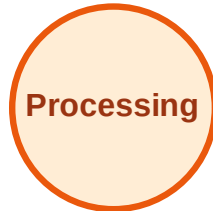
Stack (top at right): I | G | F
Visited: A, B, C, E, H, J



DFS Traversal

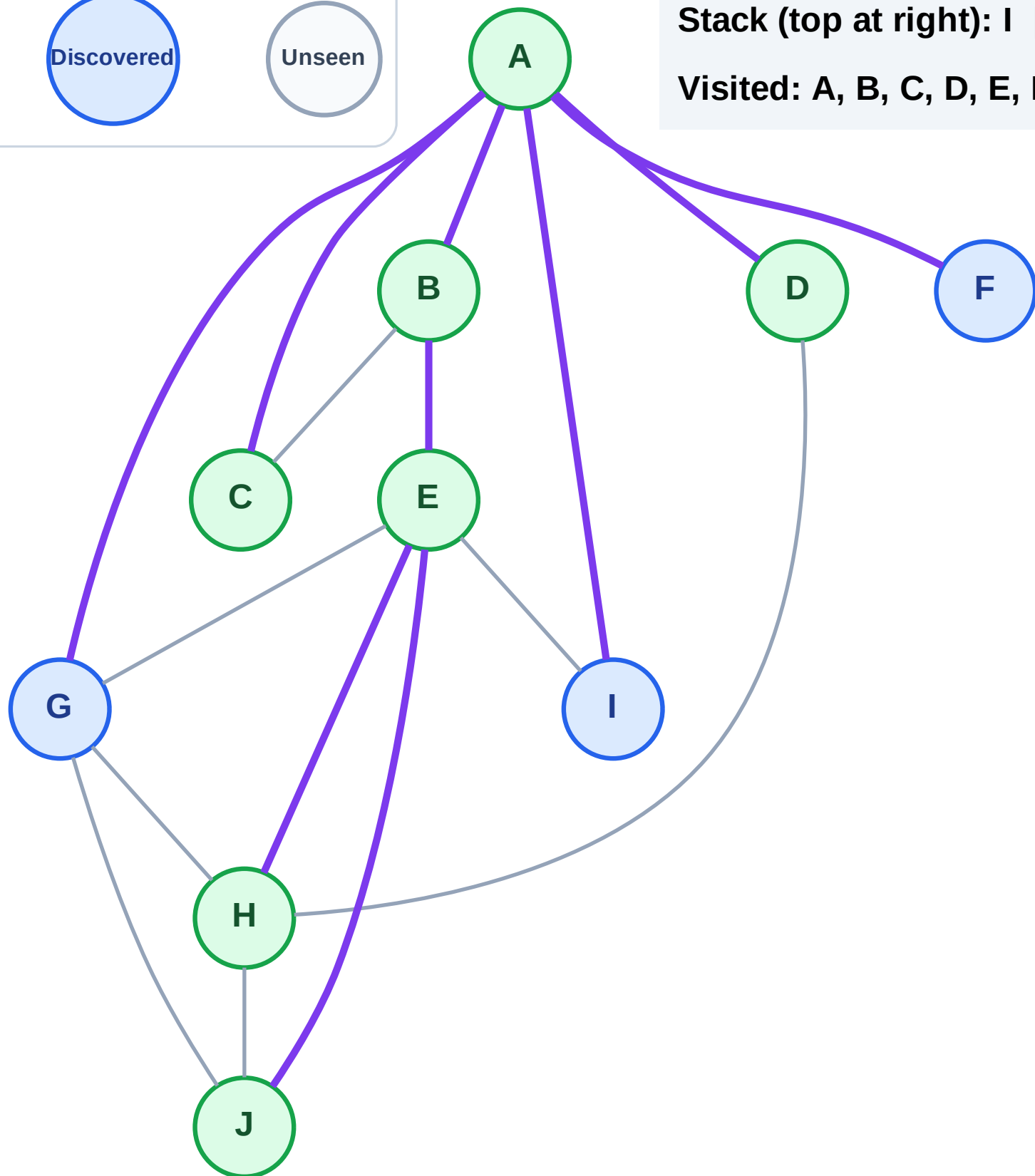
Step 15: No new neighbors from D

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node F

Legend

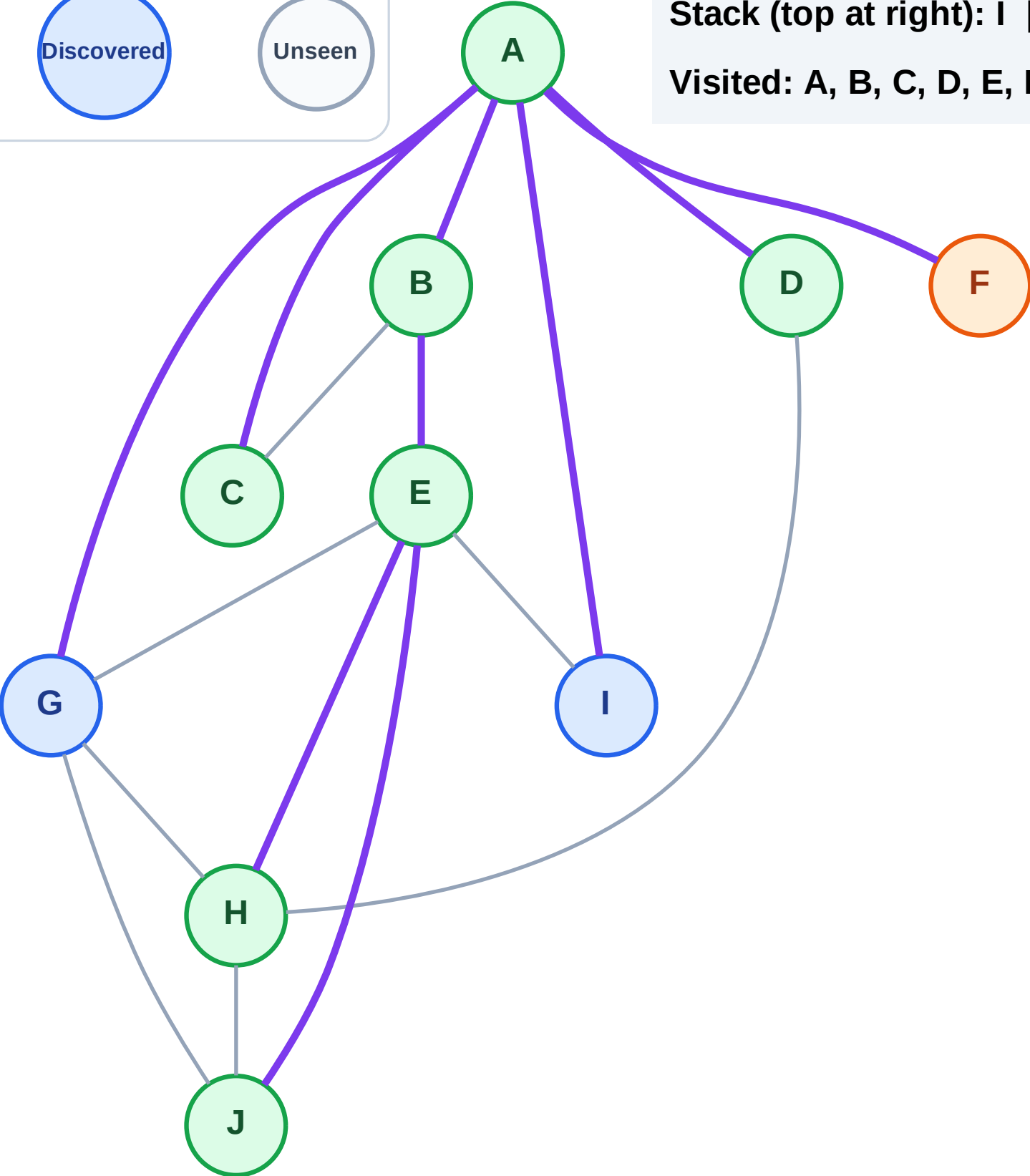
Complete

Processing

Discovered

Unseen

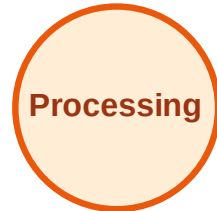
Stack (top at right): I | G
Visited: A, B, C, D, E, H, J



DFS Traversal

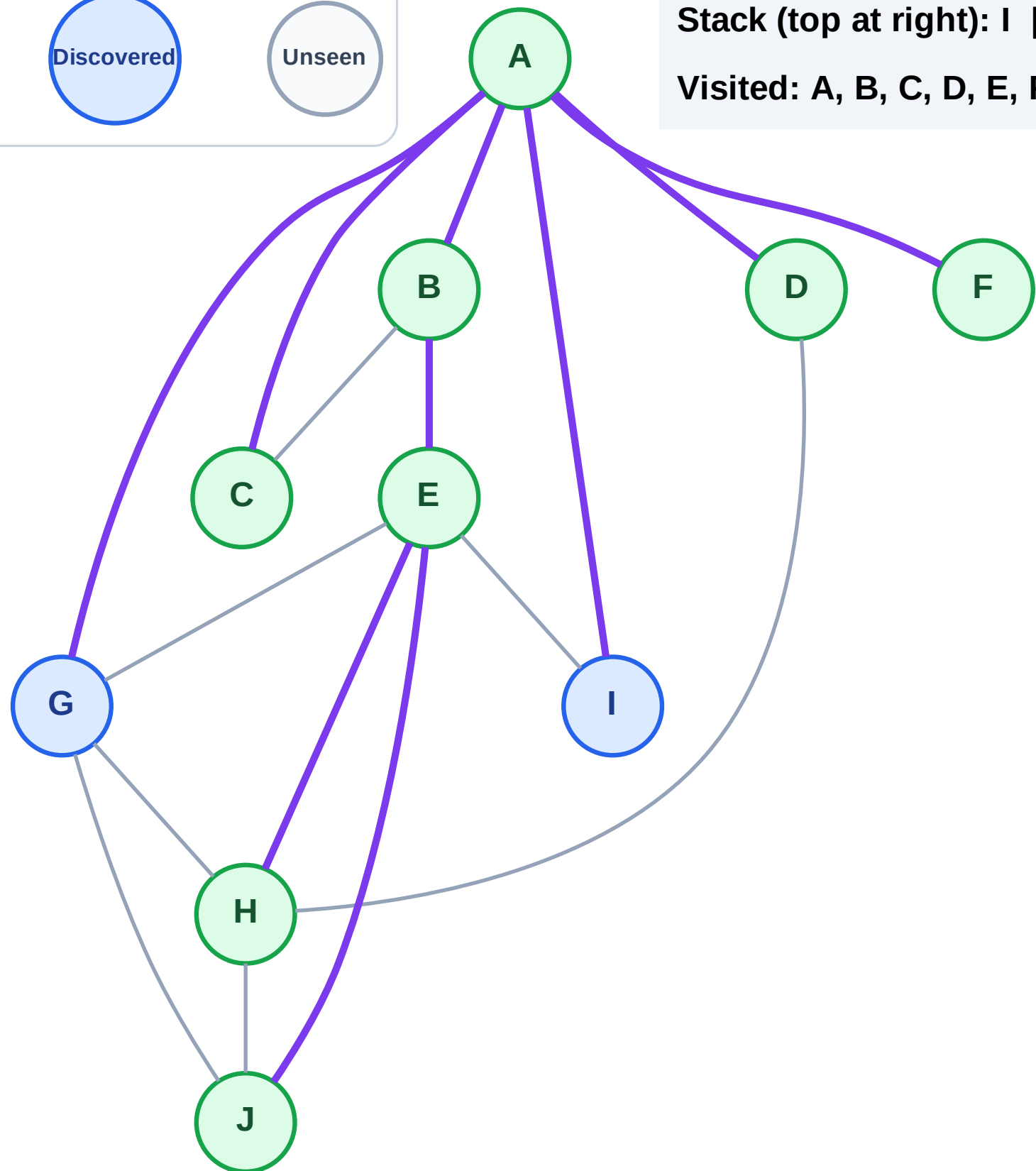
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node G

Legend

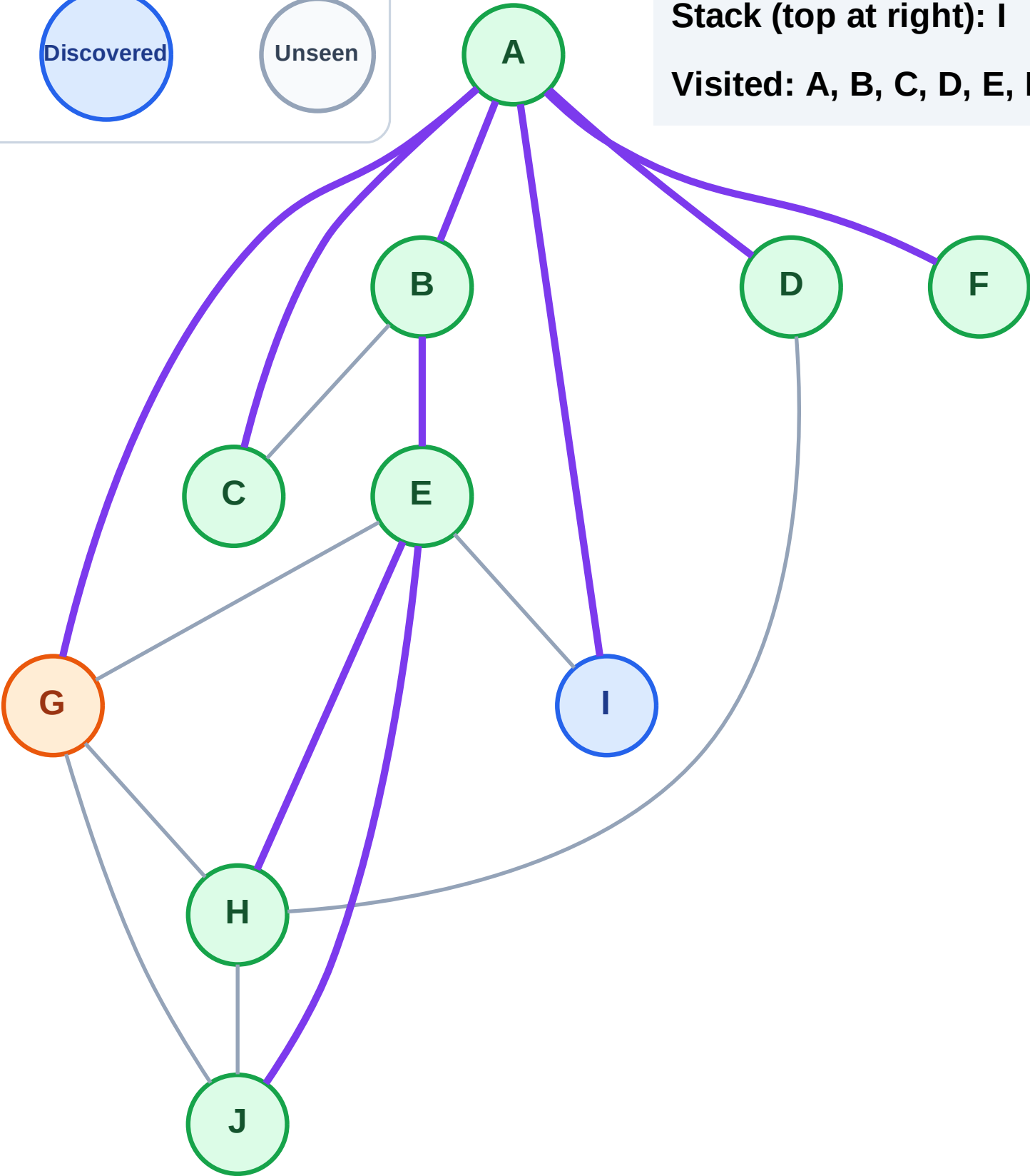
Complete

Processing

Discovered

Unseen

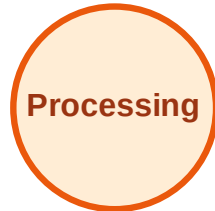
Stack (top at right): I
Visited: A, B, C, D, E, F, H, J



DFS Traversal

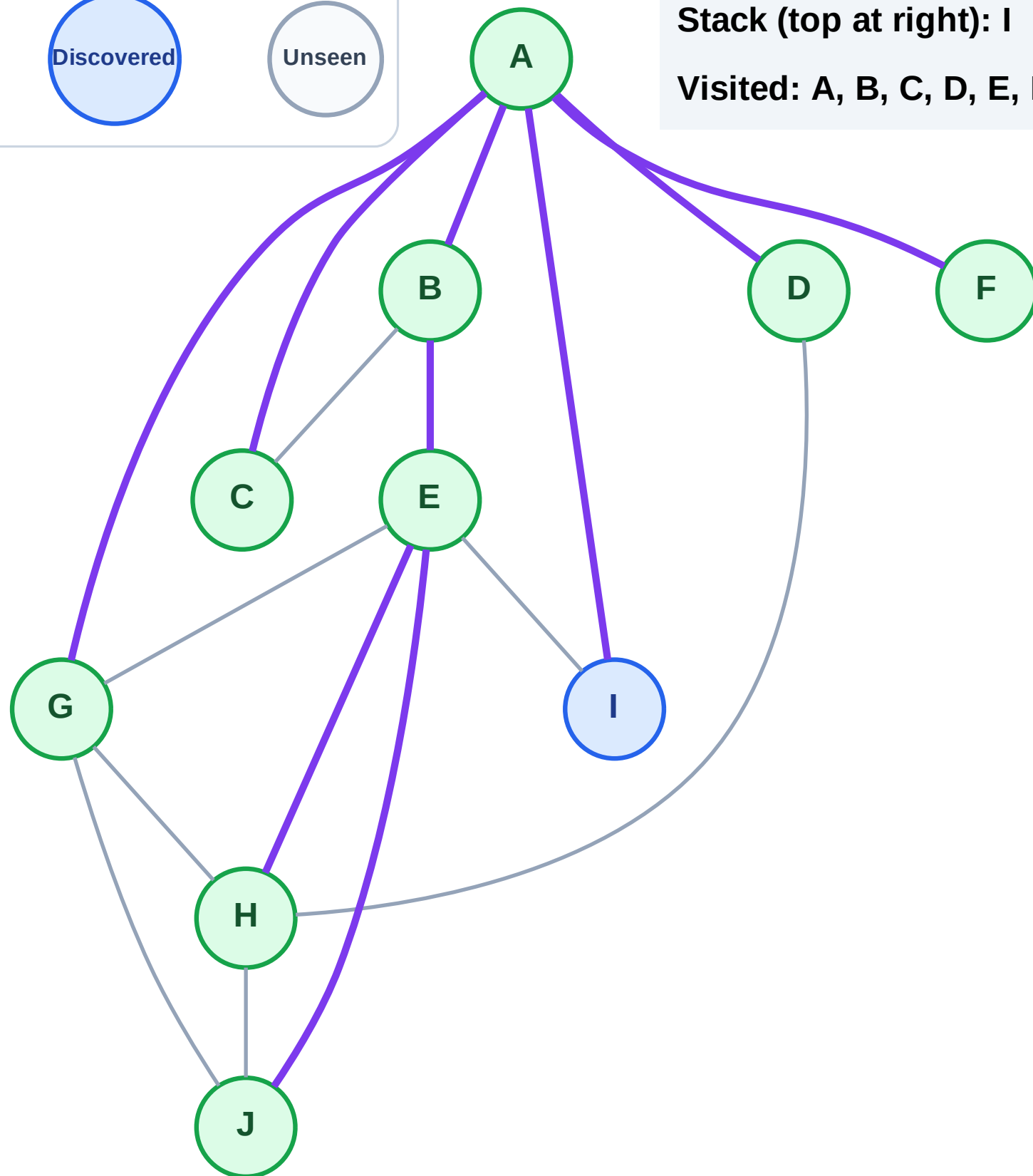
Step 19: No new neighbors from G

Legend



Stack (top at right): I

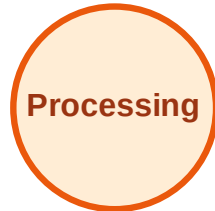
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

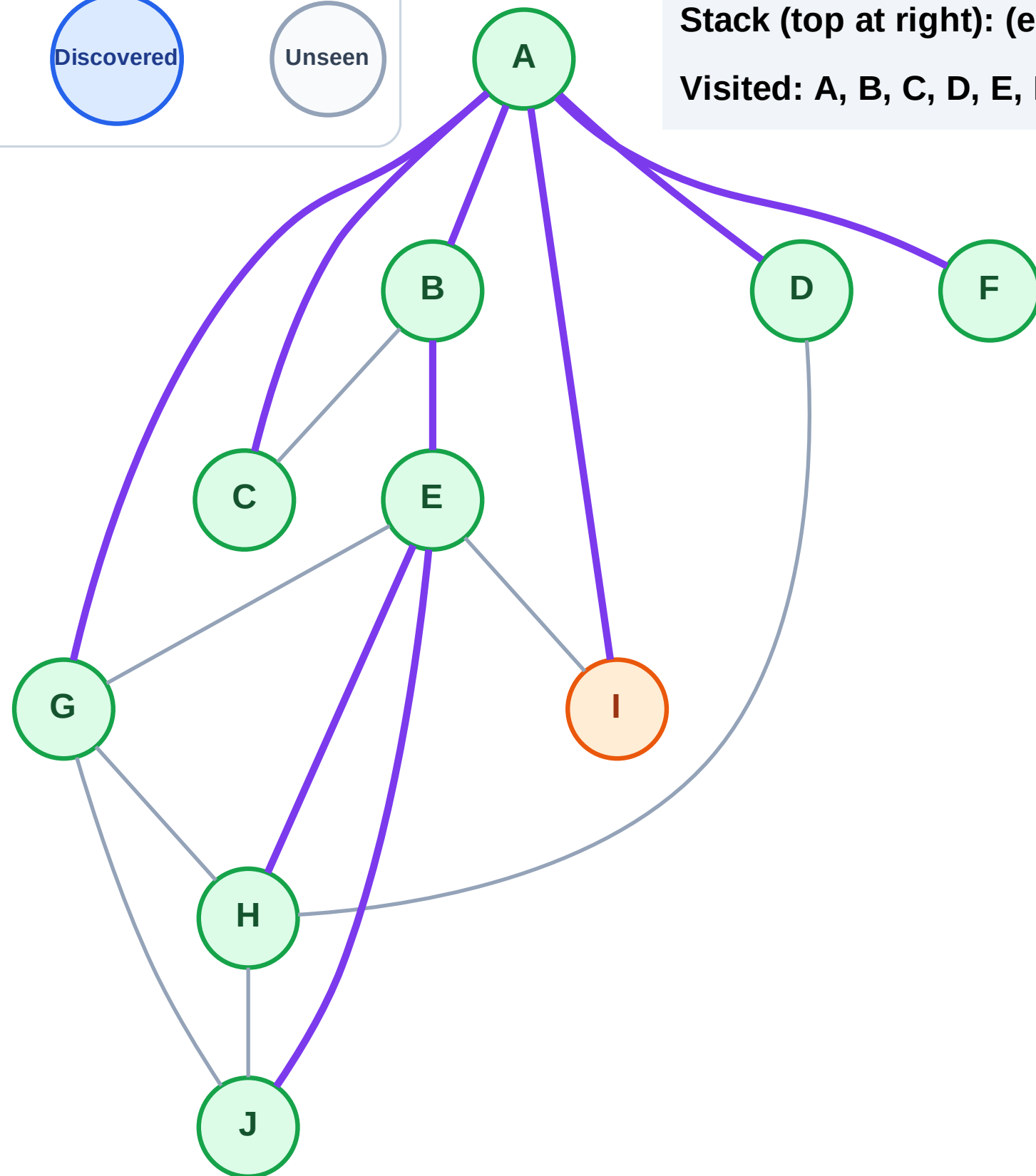
Step 20: Process node I

Legend



Stack (top at right): (empty)

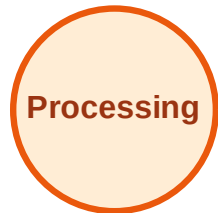
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

